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Communication devices, systems and methods

Abstract

A device includes a body, a generator configured to be releasably secured to the body, and a resilient element. The body houses a power source, a processing unit, and a bus including an electrical connector. The generator at least partially includes an impact generator element and a heat generator element, the impact generator element and the heat generator element being independently operable to convert electricity into an impact energy and a heat flux, respectively, and to communicate the impact energy and the heat flux toward an area of skin. The impact generator element includes a drive mechanism, a piston, and a tissue contact surface that is linearly actuatable along an axis. The generator further includes one or more electrical conductors engageable with the electrical connector when the generator is secured to the body to provide at least one electrical pathway from the power source to the generator.

Inventors:	Leaper; Matthew Robert (Omaha, NE)
Applicant:	DATAFEEL INC. (Washington, DC)
Family ID:	1000008592714
Assignee:	DATAFEEL INC. (Washington, DC)
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continuation parent-doc US 18092398 20230102 US 11684313 20230624 child-doc US 18143142
continuation parent-doc US 17213117 20210325 US 11589816 20230228 child-doc US 18092398
continuation parent-doc US 16757482 US 10959674 20210330 WO PCT/US2018/056814
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2088780	12/1936	Follese	N/A	N/A
2327222	12/1942	Sell	N/A	N/A
3108268	12/1962	Uttal	N/A	N/A
3594787	12/1970	Ickes	N/A	N/A
3848608	12/1973	Leonard	N/A	N/A
4167189	12/1978	Tachi	N/A	N/A
4428368	12/1983	Torii	N/A	N/A

4510939	12/1984	Brenman et al.	N/A	N/A
4535784	12/1984	Rohlicek et al.	N/A	N/A
4573481	12/1985	Bullara	N/A	N/A
4612934	12/1985	Borkan	N/A	N/A
4628933	12/1985	Michelson	N/A	N/A
4651716	12/1986	Forester et al.	N/A	N/A
4676246	12/1986	Korenaga	N/A	N/A
4837049	12/1988	Byers et al.	N/A	N/A
4885565	12/1988	Embach	N/A	N/A
4926879	12/1989	Sevrain et al.	N/A	N/A
4930504	12/1989	Diamantopoulos et al.	N/A	N/A
5024236	12/1990	Shapiro	N/A	N/A
5050587	12/1990	Sagara et al.	N/A	N/A
5109844	12/1991	De Juan, Jr. et al.	N/A	N/A
5150704	12/1991	Tatebayashi et al.	N/A	N/A
5165897	12/1991	Johnson	N/A	N/A
5175459	12/1991	Danial et al.	N/A	N/A
5215088	12/1992	Normann et al.	N/A	N/A
5259380	12/1992	Mendes et al.	N/A	N/A
5272716	12/1992	Soltz et al.	N/A	N/A
5300097	12/1993	Lerner et al.	N/A	N/A
5304207	12/1993	Stromer	N/A	N/A
5327886	12/1993	Chiu	N/A	N/A
5358503	12/1993	Bertwell et al.	N/A	N/A
5385503	12/1994	Stouffer et al.	N/A	N/A
5436622	12/1994	Gutman et al.	N/A	N/A
5443915	12/1994	Wilkie et al.	N/A	N/A
5449378	12/1994	Schouenborg	N/A	N/A
5464436	12/1994	Smith	N/A	N/A
5549660	12/1995	Mendes et al.	N/A	N/A
5551949	12/1995	Kim	N/A	N/A
5553148	12/1995	Werle	N/A	N/A
5569307	12/1995	Schulman et al.	N/A	N/A
5575761	12/1995	Hajianpour	N/A	N/A
5578065	12/1995	Hattori et al.	N/A	N/A
5616140	12/1996	Prescott	N/A	N/A
5627548	12/1996	Woo et al.	N/A	N/A
5665053	12/1996	Jacobs	N/A	N/A
5719561	12/1997	Gonzales	N/A	N/A
5776233	12/1997	Wiedemann et al.	N/A	N/A
5779483	12/1997	Cho	N/A	N/A
5800479	12/1997	Thiberg	N/A	N/A
5843074	12/1997	Cocilovo	N/A	N/A
5865771	12/1998	Shuto et al.	N/A	N/A
5876427	12/1998	Chen et al.	N/A	N/A
5895348	12/1998	Hosaka	N/A	N/A
5913883	12/1998	Alexander et al.	N/A	N/A
5935155	12/1998	Humayun et al.	N/A	N/A
5944748	12/1998	Mager et al.	N/A	N/A
5948006	12/1998	Mann	N/A	N/A

5957569	12/1998	Helbig et al.	N/A	N/A
5957960	12/1998	Chen et al.	N/A	N/A
6038485	12/1999	Axelgaard	N/A	N/A
6045575	12/1999	Rosen et al.	N/A	N/A
6063108	12/1999	Salansky et al.	N/A	N/A
6074411	12/1999	Lai et al.	N/A	N/A
6096066	12/1999	Chen et al.	N/A	N/A
6099554	12/1999	Nordquist et al.	N/A	N/A
6104820	12/1999	Soza	N/A	N/A
6107466	12/1999	Hasan et al.	N/A	N/A
6156028	12/1999	Prescott	N/A	N/A
6228103	12/2000	Grey et al.	N/A	N/A
6275213	12/2000	Tremblay et al.	N/A	N/A
6277085	12/2000	Flynn	N/A	N/A
6400989	12/2001	Eckmiller	N/A	N/A
6402739	12/2001	Neev	N/A	N/A
6413230	12/2001	Haupt et al.	N/A	N/A
6458157	12/2001	Suaning	N/A	N/A
6547788	12/2002	Maguire et al.	N/A	N/A
6712613	12/2003	Depta	N/A	N/A
6735480	12/2003	Giuntoli et al.	N/A	N/A
6819312	12/2003	Fish	N/A	N/A
6889090	12/2004	Kreindel	N/A	N/A
7013179	12/2005	Carter et al.	N/A	N/A
7077858	12/2005	Fletcher et al.	N/A	N/A
7081128	12/2005	Hart et al.	N/A	N/A
7110820	12/2005	Tcheng et al.	N/A	N/A
7113177	12/2005	Franzen	N/A	N/A
7182739	12/2006	Kopanic et al.	N/A	N/A
7238162	12/2006	Dehli	N/A	N/A
7241291	12/2006	Kreindel et al.	N/A	N/A
7324094	12/2007	Moilanen et al.	N/A	N/A
7336266	12/2007	Hayward et al.	N/A	N/A
7384405	12/2007	Rhoades	N/A	N/A
7390157	12/2007	Kramer	N/A	N/A
7615015	12/2008	Coleman	N/A	N/A
7687678	12/2009	Jacobs	N/A	N/A
7696860	12/2009	Gilson et al.	N/A	N/A
7722655	12/2009	Lee	N/A	N/A
7771374	12/2009	Slatkine	N/A	N/A
7818061	12/2009	Palmer	N/A	N/A
7825903	12/2009	Anastas et al.	N/A	N/A
7988649	12/2010	Kost	N/A	N/A
8013847	12/2010	Anastas	N/A	N/A
8027491	12/2010	LeDonne	N/A	N/A
8139803	12/2011	Afshar	N/A	N/A
8147533	12/2011	Baxter et al.	N/A	N/A
8157753	12/2011	Nichols	N/A	N/A
8170656	12/2011	Tan et al.	N/A	N/A
8175718	12/2011	Wahlgren et al.	N/A	N/A

8203529	12/2011	Rogowitz et al.	N/A	N/A
8308558	12/2011	Thorner	N/A	N/A
8316166	12/2011	Grant et al.	N/A	N/A
8362882	12/2012	Heubel et al.	N/A	N/A
8364257	12/2012	Van Den Eerenbeemd et al.	N/A	N/A
8369944	12/2012	Macho et al.	N/A	N/A
8506506	12/2012	Nebrigic et al.	N/A	N/A
8552847	12/2012	Hill	N/A	N/A
8632576	12/2013	Quisenberry	N/A	N/A
8633907	12/2013	Mahalingam	N/A	N/A
8686951	12/2013	Vartanian et al.	N/A	N/A
8696357	12/2013	Aldossary	N/A	N/A
8702769	12/2013	Eckhouse et al.	N/A	N/A
8711118	12/2013	Short et al.	N/A	N/A
8740960	12/2013	Baxter et al.	N/A	N/A
8766786	12/2013	Radivojevic	N/A	N/A
8767996	12/2013	Lin et al.	N/A	N/A
8906009	12/2013	Nebrigic et al.	N/A	N/A
8917167	12/2013	Selker	N/A	N/A
8922503	12/2013	Ciesla et al.	N/A	N/A
8945104	12/2014	Boone, III et al.	N/A	N/A
8965500	12/2014	Macho et al.	N/A	N/A
8979775	12/2014	Schafer et al.	N/A	N/A
9008794	12/2014	Alexandre	N/A	N/A
9024874	12/2014	Stetten et al.	N/A	N/A
9042971	12/2014	Brumback et al.	N/A	N/A
9046919	12/2014	Niknejad	N/A	N/A
9050133	12/2014	Boone, III et al.	N/A	N/A
9053617	12/2014	Ramstein et al.	N/A	N/A
9064387	12/2014	Bhatia et al.	N/A	N/A
9078065	12/2014	Karam et al.	N/A	N/A
9083821	12/2014	Hughes	N/A	N/A
9092953	12/2014	Mortimer et al.	N/A	N/A
9092954	12/2014	Visitacion et al.	N/A	N/A
9095359	12/2014	Behnke, II et al.	N/A	N/A
9170650	12/2014	Ramstein et al.	N/A	N/A
9195350	12/2014	Radivojevic et al.	N/A	N/A
9198792	12/2014	Skahan et al.	N/A	N/A
9213408	12/2014	Gandhi et al.	N/A	N/A
9254382	12/2015	Ahmad et al.	N/A	N/A
9272141	12/2015	Nichols	N/A	N/A
9274603	12/2015	Modarres et al.	N/A	N/A
9282893	12/2015	Longinotti-Buitoni et al.	N/A	N/A
9295607	12/2015	Rosenberg	N/A	N/A
9308363	12/2015	Goroszeniuk et al.	N/A	N/A
9311792	12/2015	Kosonen et al.	N/A	N/A
9314650	12/2015	Rosenberg et al.	N/A	N/A
9333144	12/2015	Baxter et al.	N/A	N/A

9333377	12/2015	Rosenberg	N/A	N/A
9364667	12/2015	Dinsmoor et al.	N/A	N/A
9375345	12/2015	Levinson et al.	N/A	N/A
9407130	12/2015	Garcia et al.	N/A	N/A
9408316	12/2015	Bailey et al.	N/A	N/A
9417694	12/2015	Birnbaum et al.	N/A	N/A
9452287	12/2015	Rosenbluth et al.	N/A	N/A
9454915	12/2015	Aldossary et al.	N/A	N/A
9466187	12/2015	Grant et al.	N/A	N/A
9468847	12/2015	Bekri	N/A	N/A
9504826	12/2015	Flyash et al.	N/A	N/A
9553625	12/2016	Hatanaka et al.	N/A	N/A
9561357	12/2016	Hall et al.	N/A	N/A
9582034	12/2016	Von Badinski et al.	N/A	N/A
9582035	12/2016	Connor	N/A	N/A
9582072	12/2016	Connor	N/A	N/A
9585687	12/2016	Tenenbaum et al.	N/A	N/A
9607492	12/2016	Churovich	N/A	N/A
9619034	12/2016	Birnbaum et al.	N/A	N/A
9623907	12/2016	Marti et al.	N/A	N/A
9626845	12/2016	Eagleman et al.	N/A	N/A
9652946	12/2016	Ramstein et al.	N/A	N/A
9669233	12/2016	Quisenberry et al.	N/A	N/A
9672701	12/2016	Evreinov et al.	N/A	N/A
9672791	12/2016	Kapinos et al.	N/A	N/A
9703381	12/2016	Gallo et al.	N/A	N/A
9754078	12/2016	Ramsay et al.	N/A	N/A
9754464	12/2016	Sinkov	N/A	N/A
9829977	12/2016	Heubel et al.	N/A	N/A
9884179	12/2017	Bouton et al.	N/A	N/A
9898085	12/2017	Saboune et al.	N/A	N/A
9936900	12/2017	Chang et al.	N/A	N/A
9940811	12/2017	Chang et al.	N/A	N/A
9943698	12/2017	Chase et al.	N/A	N/A
9950147	12/2017	Mehta	N/A	N/A
9961435	12/2017	Goyal et al.	N/A	N/A
10019912	12/2017	Eagleman et al.	N/A	N/A
10039928	12/2017	Hyde et al.	N/A	N/A
10058476	12/2017	Baxter et al.	N/A	N/A
10070799	12/2017	Ang et al.	N/A	N/A
10080906	12/2017	Schwarz	N/A	N/A
10111010	12/2017	Alexiou et al.	N/A	N/A
10123937	12/2017	Pisharodi et al.	N/A	N/A
10181331	12/2018	Eagleman et al.	N/A	N/A
10198076	12/2018	Eagleman et al.	N/A	N/A
10200332	12/2018	Wu et al.	N/A	N/A
10216278	12/2018	Nakamura et al.	N/A	N/A
10234934	12/2018	Connor	N/A	N/A
10255771	12/2018	Baron et al.	N/A	N/A
10275029	12/2018	Jones et al.	N/A	N/A

10285902	12/2018	Pamplin et al.	N/A	N/A
10321873	12/2018	Connor	N/A	N/A
10371544	12/2018	Yoo et al.	N/A	N/A
10384076	12/2018	Wagenaar Cacciola et al.	N/A	N/A
10437340	12/2018	Sullivan	N/A	N/A
10456604	12/2018	Cheatham, III et al.	N/A	N/A
10524978	12/2019	Marton et al.	N/A	N/A
10589087	12/2019	Tyler et al.	N/A	N/A
10625074	12/2019	Rosenbluth et al.	N/A	N/A
10699538	12/2019	Novich et al.	N/A	N/A
10748448	12/2019	Knott et al.	N/A	N/A
10888708	12/2020	Stephens	N/A	N/A
10945878	12/2020	Deng et al.	N/A	N/A
11247039	12/2021	Schwarz et al.	N/A	N/A
11253717	12/2021	Schwarz et al.	N/A	N/A
11439841	12/2021	Zimmerman	N/A	N/A
11464993	12/2021	Schwarz et al.	N/A	N/A
11701253	12/2022	Black et al.	N/A	N/A
11712367	12/2022	Simons et al.	N/A	N/A
2001/0043847	12/2000	Kramer	N/A	N/A
2002/0143373	12/2001	Courtnage et al.	N/A	N/A
2003/0023297	12/2002	Byers et al.	N/A	N/A
2003/0032900	12/2002	Ella	N/A	N/A
2003/0083599	12/2002	Kitov	N/A	N/A
2003/0153961	12/2002	Babaev	N/A	N/A
2003/0181116	12/2002	Van Heerden et al.	N/A	N/A
2003/0187488	12/2002	Kreindel et al.	N/A	N/A
2003/0212350	12/2002	Tadlock	N/A	N/A
2004/0039254	12/2003	Stivoric et al.	N/A	N/A
2004/0040800	12/2003	Anastas et al.	N/A	N/A
2004/0082886	12/2003	Timpson	N/A	N/A
2004/0210122	12/2003	Sieburg	N/A	N/A
2004/0241623	12/2003	Lenay et al.	N/A	N/A
2004/0260211	12/2003	Maalouf	N/A	N/A
2005/0078846	12/2004	Single	N/A	N/A
2005/0177093	12/2004	Barry et al.	N/A	N/A
2005/0209537	12/2004	Gleason et al.	N/A	N/A
2006/0036201	12/2005	Cohen	N/A	N/A
2006/0041207	12/2005	Gross	N/A	N/A
2006/0052695	12/2005	Adam	N/A	N/A
2006/0116611	12/2005	Richter	N/A	N/A
2006/0135890	12/2005	Tsai	N/A	N/A
2006/0247754	12/2005	Greenberg et al.	N/A	N/A
2006/0253051	12/2005	Milne et al.	N/A	N/A
2006/0264926	12/2005	Kochamba	N/A	N/A
2007/0010810	12/2006	Kochamba	N/A	N/A
2007/0016425	12/2006	Ward	N/A	N/A
2007/0027411	12/2006	Ella et al.	N/A	N/A
2007/0063849	12/2006	Rosella et al.	N/A	N/A

2007/0255187	12/2006	Branch	N/A	N/A
2007/0255355	12/2006	Altshuler et al.	N/A	N/A
2007/0282400	12/2006	Gorham	N/A	N/A
2008/0014011	12/2007	Rossen	N/A	N/A
2008/0023597	12/2007	Wyner et al.	N/A	N/A
2008/0188911	12/2007	Chao	N/A	N/A
2008/0195000	12/2007	Spooner et al.	N/A	N/A
2008/0215039	12/2007	Slatkine et al.	N/A	N/A
2008/0312647	12/2007	Knopp et al.	N/A	N/A
2009/0048547	12/2008	Chen	N/A	N/A
2009/0076421	12/2008	Grant, Jr.	N/A	N/A
2009/0098519	12/2008	Byerly	N/A	N/A
2009/0112134	12/2008	Avni	N/A	N/A
2009/0120105	12/2008	Ramsay et al.	N/A	N/A
2009/0131840	12/2008	Lee	N/A	N/A
2009/0143706	12/2008	Acosta	N/A	N/A
2009/0156958	12/2008	Mehta et al.	N/A	N/A
2009/0171251	12/2008	Rybyanets et al.	N/A	N/A
2009/0180646	12/2008	Vulfson et al.	N/A	N/A
2009/0221938	12/2008	Rosenberg et al.	N/A	N/A
2010/0004536	12/2009	Rosenberg	N/A	N/A
2010/0016761	12/2009	Rosenberg	N/A	N/A
2010/0042020	12/2009	Ben-Ezra	N/A	N/A
2010/0069800	12/2009	Hsu	N/A	N/A
2010/0073150	12/2009	Olson et al.	N/A	N/A
2010/0134327	12/2009	Dinh et al.	N/A	N/A
2010/0137752	12/2009	Heine et al.	N/A	N/A
2010/0141407	12/2009	Heubel et al.	N/A	N/A
2010/0145242	12/2009	Tsai	N/A	N/A
2010/0145425	12/2009	Jung et al.	N/A	N/A
2010/0162109	12/2009	Chatterjee et al.	N/A	N/A
2010/0204619	12/2009	Rosenberg	N/A	N/A
2010/0204694	12/2009	Mehta et al.	N/A	N/A
2010/0259472	12/2009	Radivojevic et al.	N/A	N/A
2010/0292746	12/2009	Gorham	N/A	N/A
2010/0298745	12/2009	Liu et al.	N/A	N/A
2010/0304864	12/2009	Johnson et al.	N/A	N/A
2010/0305495	12/2009	Anderson et al.	N/A	N/A
2011/0004261	12/2010	Sham et al.	N/A	N/A
2011/0040235	12/2010	Castel	N/A	N/A
2011/0046523	12/2010	Altshuler et al.	N/A	N/A
2011/0148607	12/2010	Zeleny	N/A	N/A
2011/0166559	12/2010	Eckhouse et al.	N/A	N/A
2011/0270140	12/2010	Israeli	N/A	N/A
2012/0010603	12/2011	Milner et al.	N/A	N/A
2012/0023785	12/2011	Barnes et al.	N/A	N/A
2012/0035513	12/2011	Afshar	N/A	N/A
2012/0070805	12/2011	Wong et al.	N/A	N/A
2012/0123304	12/2011	Rybyanets	N/A	N/A
2012/0232780	12/2011	Delson et al.	N/A	N/A

2012/0253236	12/2011	Snow et al.	N/A	N/A
2012/0253416	12/2011	Erez et al.	N/A	N/A
2012/0277587	12/2011	Adanny et al.	N/A	N/A
2012/0316381	12/2011	Teggatz et al.	N/A	N/A
2013/0041296	12/2012	Tass et al.	N/A	N/A
2013/0046212	12/2012	Nichols	N/A	N/A
2013/0131559	12/2012	Vandenbelt et al.	N/A	N/A
2013/0178764	12/2012	Eckhouse et al.	N/A	N/A
2013/0184693	12/2012	Neev	N/A	N/A
2013/0202674	12/2012	Ericson	N/A	N/A
2013/0204169	12/2012	Poepperling et al.	N/A	N/A
2013/0218456	12/2012	Zelek et al.	N/A	N/A
2013/0331913	12/2012	Levi et al.	N/A	N/A
2014/0007957	12/2013	Vendramini	N/A	N/A
2014/0036053	12/2013	Clingman et al.	N/A	N/A
2014/0070957	12/2013	Longinotti-Buitoni et al.	N/A	N/A
2014/0085077	12/2013	Luna et al.	N/A	N/A
2014/0128780	12/2013	Kennedy et al.	N/A	N/A
2014/0143737	12/2013	Mistry et al.	N/A	N/A
2014/0163439	12/2013	Uryash et al.	N/A	N/A
2014/0172062	12/2013	Yoon	N/A	N/A
2014/0180181	12/2013	Von Oepen et al.	N/A	N/A
2014/0184384	12/2013	Zhu et al.	N/A	N/A
2014/0198034	12/2013	Bailey et al.	N/A	N/A
2014/0198035	12/2013	Bailey et al.	N/A	N/A
2014/0206947	12/2013	Isserow et al.	N/A	N/A
2014/0206976	12/2013	Thompson et al.	N/A	N/A
2014/0214206	12/2013	Steinberg et al.	N/A	N/A
2014/0221879	12/2013	Chu	N/A	N/A
2014/0266570	12/2013	Sharma et al.	N/A	N/A
2014/0266571	12/2013	Sharma et al.	N/A	N/A
2014/0276271	12/2013	Stryker et al.	N/A	N/A
2014/0326241	12/2013	Martin et al.	N/A	N/A
2015/0038886	12/2014	Snow	N/A	N/A
2015/0045702	12/2014	Lin	N/A	N/A
2015/0105129	12/2014	Chapman	N/A	N/A
2015/0119771	12/2014	Roberts	N/A	N/A
2015/0182762	12/2014	Schafer et al.	N/A	N/A
2015/0190607	12/2014	Sugio et al.	N/A	N/A
2015/0201181	12/2014	Moore et al.	N/A	N/A
2015/0217142	12/2014	Schafer	N/A	N/A
2015/0220199	12/2014	Wang et al.	N/A	N/A
2015/0235529	12/2014	Deschamps	N/A	N/A
2015/0257970	12/2014	Mücke et al.	N/A	N/A
2015/0265214	12/2014	De Kok et al.	N/A	N/A
2015/0283022	12/2014	Lee et al.	N/A	N/A
2015/0283026	12/2014	Rosenberg	N/A	N/A
2015/0283277	12/2014	Schafer et al.	N/A	N/A
2015/0290454	12/2014	Tyler et al.	N/A	N/A

2015/0305969	12/2014	Giraud et al.	N/A	N/A
2015/0306373	12/2014	Bouton et al.	N/A	N/A
2015/0321000	12/2014	Rosenbluth et al.	N/A	N/A
2015/0335288	12/2014	Toth et al.	N/A	N/A
2015/0339899	12/2014	Ozaki et al.	N/A	N/A
2015/0351690	12/2014	Toth et al.	N/A	N/A
2015/0364018	12/2014	Mirov et al.	N/A	N/A
2015/0366504	12/2014	Connor	N/A	N/A
2016/0012688	12/2015	Eagleman et al.	N/A	N/A
2016/0012689	12/2015	Evreinov et al.	N/A	N/A
2016/0018920	12/2015	Deokar et al.	N/A	N/A
2016/0058156	12/2015	Chiasson	N/A	N/A
2016/0095789	12/2015	Baxter et al.	N/A	N/A
2016/0101010	12/2015	Hsu	N/A	N/A
2016/0129248	12/2015	Creasey et al.	N/A	N/A
2016/0136040	12/2015	Li	N/A	N/A
2016/0138144	12/2015	Olsérius et al.	N/A	N/A
2016/0165965	12/2015	Ellis et al.	N/A	N/A
2016/0187977	12/2015	Cruz-Hernandez et al.	N/A	N/A
2016/0195928	12/2015	Wagner et al.	N/A	N/A
2016/0235980	12/2015	Berman et al.	N/A	N/A
2016/0235983	12/2015	Berman et al.	N/A	N/A
2016/0246378	12/2015	Sampanes et al.	N/A	N/A
2016/0255733	12/2015	Jung et al.	N/A	N/A
2016/0267344	12/2015	Yamamoto	N/A	N/A
2016/0302721	12/2015	Wiedenhoefer et al.	N/A	N/A
2016/0310353	12/2015	Barasch	N/A	N/A
2016/0338644	12/2015	Connor	N/A	N/A
2016/0346153	12/2015	Hodges, IV	N/A	N/A
2016/0346162	12/2015	Powers	N/A	N/A
2016/0349790	12/2015	Connor	N/A	N/A
2016/0360974	12/2015	Lange	N/A	N/A
2016/0374886	12/2015	Wyatt et al.	N/A	N/A
2017/0004685	12/2016	Karsten	N/A	N/A
2017/0011602	12/2016	Brav et al.	N/A	N/A
2017/0036002	12/2016	Ignon et al.	N/A	N/A
2017/0043150	12/2016	Kim	N/A	N/A
2017/0080255	12/2016	Law et al.	N/A	N/A
2017/0095692	12/2016	Chang et al.	N/A	N/A
2017/0095693	12/2016	Chang et al.	N/A	N/A
2017/0098350	12/2016	Ebeling et al.	N/A	N/A
2017/0156662	12/2016	Goodall et al.	N/A	N/A
2017/0165486	12/2016	Harry et al.	N/A	N/A
2017/0188894	12/2016	Chang et al.	N/A	N/A
2017/0209708	12/2016	Schwarz	N/A	N/A
2017/0221323	12/2016	Nakamura et al.	N/A	N/A
2017/0246076	12/2016	Miller et al.	N/A	N/A
2017/0249810	12/2016	Zerick et al.	N/A	N/A
2017/0252534	12/2016	Nofzinger	N/A	N/A
2017/0273601	12/2016	Wang et al.	N/A	N/A

2017/0304145	12/2016	Pepe	N/A	N/A
2017/0304646	12/2016	Pryor et al.	N/A	N/A
2017/0348049	12/2016	Vrba et al.	N/A	N/A
2018/0015299	12/2017	Kawa	N/A	N/A
2018/0021579	12/2017	Kahana et al.	N/A	N/A
2018/0081439	12/2017	Daniels	N/A	N/A
2018/0099143	12/2017	Kim et al.	N/A	N/A
2018/0207056	12/2017	Howard et al.	N/A	N/A
2018/0243161	12/2017	Lenke	N/A	N/A
2018/0296166	12/2017	LeBoeuf et al.	N/A	N/A
2018/0303702	12/2017	Novich et al.	N/A	N/A
2018/0307314	12/2017	Connor	N/A	N/A
2019/0029878	12/2018	Linder et al.	N/A	N/A
2019/0064924	12/2018	Nocon	N/A	N/A
2019/0076643	12/2018	Siegle et al.	N/A	N/A
2019/0105101	12/2018	Narisawa	N/A	N/A
2019/0125262	12/2018	Markel	N/A	N/A
2019/0151604	12/2018	Harper et al.	N/A	N/A
2019/0183388	12/2018	Cohen et al.	N/A	N/A
2019/0188976	12/2018	Aleksov et al.	N/A	N/A
2019/0278374	12/2018	Sato et al.	N/A	N/A
2019/0343205	12/2018	Clausen	N/A	N/A
2019/0350752	12/2018	Aguiar et al.	N/A	N/A
2020/0163572	12/2019	Mann	N/A	N/A
2020/0174583	12/2019	Wang	N/A	N/A
2020/0186104	12/2019	Honda et al.	N/A	N/A
2020/0245931	12/2019	Chmelik	N/A	N/A
2020/0246520	12/2019	Chang	N/A	N/A
2020/0253485	12/2019	Kang et al.	N/A	N/A
2020/0276079	12/2019	Cheng	N/A	A61H 23/0263
2021/0286180	12/2020	Samec et al.	N/A	N/A
2022/0347010	12/2021	Kerth et al.	N/A	N/A
2023/0185376	12/2022	Sembaluk et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2015202395	12/2014	AU	N/A
103550029	12/2013	CN	N/A
105726286	12/2017	CN	N/A
108938359	12/2017	CN	N/A
210228475	12/2019	CN	N/A
0556999	12/1992	EP	N/A
2203797	12/2009	EP	N/A
2450075	12/2011	EP	N/A
2203797	12/2013	EP	N/A
2506773	12/2017	EP	N/A
3003473	12/2017	EP	N/A
2790088	12/2018	EP	N/A
H059543	12/1992	JP	N/A
4143114	12/2007	JP	N/A

4744968	12/2010	JP	N/A
2017035414	12/2016	JP	N/A
6178566	12/2016	JP	N/A
6652164	12/2019	JP	N/A
2022050711	12/2021	JP	N/A
930001844	12/1992	KR	N/A
20010008111	12/2000	KR	N/A
20040042912	12/2003	KR	N/A
200395260	12/2004	KR	N/A
200412070	12/2005	KR	N/A
100746214	12/2006	KR	N/A
100787874	12/2006	KR	N/A
100793861	12/2007	KR	N/A
101064115	12/2010	KR	N/A
20110119468	12/2010	KR	N/A
20120018245	12/2011	KR	N/A
101123926	12/2011	KR	N/A
20120128427	12/2011	KR	N/A
101315403	12/2012	KR	N/A
101319429	12/2012	KR	N/A
101367893	12/2013	KR	N/A
101767485	12/2016	KR	N/A
181413	12/2011	SG	N/A
0240095	12/2001	WO	N/A
2004023235	12/2003	WO	N/A
2012074483	12/2011	WO	N/A
2013118122	12/2012	WO	N/A
2014117125	12/2013	WO	N/A
2014194200	12/2013	WO	N/A
2015083183	12/2014	WO	N/A
2016019250	12/2015	WO	N/A
2016201366	12/2015	WO	N/A
2017158583	12/2016	WO	N/A
2017173436	12/2016	WO	N/A
2018013835	12/2017	WO	N/A
2018013835	12/2017	WO	N/A
2018048907	12/2017	WO	N/A
2018143418	12/2017	WO	N/A
2020219350	12/2019	WO	N/A

OTHER PUBLICATIONS

Ashley Carman, “This ring will vibrate in coordination with your loved one's pulse; They're alive! And not answering your texts!,” Circuit Breaker, The Verge, Aug. 10, 2016, <https://www.theverge.com/circuitbreaker/2016/8/10/12423956/thetouch-hb-ring-heartrate-pulse>. cited by applicant

Eagleman D., “Can We Create New Senses for Humans?,” Transcript of TED Talk, (Last Accessed May 17, 2022), Retrieved from URL:

https://www.ted.com/talks/david_eagleman_can_we_create_new_senses_for_humans/transcript?language=enE. cited by applicant

Extended European Search Report for European Application No. 18871368.9, mailed Sep. 22, 2021, 8

Pages. cited by applicant

International Preliminary Report on Patentability for International Application No.

PCT/US2018/056814, dated Mar. 23, 2020, 50 Pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2018/056814,

mailed Feb. 22, 2019, 21 Pages. cited by applicant

Iwamoto et al., "Non-contact Method for Producing Tactile Sensation Using Airborne Ultrasound", Eurohaptics 2008, LNCS 5024, pp. 504-513, http://www.researchgate.net/publication/221011909_Non-contact_Method_for_Producing_Tactile_Sensation_Using_Airborne_Ultrasound. cited by applicant

Maereg et al., "Wearable Vibrotactile Haptic Device for Stiffness Discrimination during Virtual Interactions," Journal of Frontiers in Robotics and AI, vol. 4, 2017,

<https://www.frontiersin.org/article/10.3389/frobt.2017.00042>. cited by applicant

Novich et al., "Using space and time to encode vibrotactile information: toward an estimate of the skin's achievable throughput," Experimental Brain Research. 233, p. 2777-2788(2015). available at:

<Http://link.springer.com/article/10.1007/s00221-015-4346-1>. cited by applicant

Seneviratne et al., "A Survey of Wearable Devices and Challenges," Article, IEEE Communications Surveys & tutorials, Jul. 2017, (52 pages), (99)1-1, available at:

http://www.researchgate.net/publications/318717275_A_Survey_of_Wearable_Devices_and_Challenges. cited by applicant

Shull et al., "Haptic wearables as sensory replacement, sensory augmentation and trainer a review," Journal of NeuroEngineering and Rehabilitation, vol. 12, Article #59, 2015,

<https://doi.org/10.1186/s12984-015-0055-z>. cited by applicant

Singhal, Anshul et al., Perceptual Interactions in thermo-tactile displays, Jun. 2017, 5 pages, 2017

Institute of Electrical and Electronics Engineers (IEEE) World Haptics Conference, Munich, Germany, Jun. 6-9, 2017. cited by applicant

Petition for Post-Grant Review of U.S. Pat. No. 12,036,174 Case No. PGR2025-00026, 2025. cited by applicant

Declaration of Morten O. Jensen, PhD, DrMed in Support of Petition for Post Grant Review of U.S. Pat. No. 12,036,174 Case No. PGR2025-00026, 2025. cited by applicant

Primary Examiner: Gaworecki; Mark R

Attorney, Agent or Firm: POLSINELLI PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/958,622 filed Nov. 25, 2024, which is a continuation of U.S. patent application Ser. No. 18/956,924 filed Nov. 22, 2024, which is a continuation of U.S. patent application Ser. No. 18/893,913, filed Sep. 23, 2024, now U.S. Pat. No. 12,193,840, which is a continuation of U.S. patent application Ser. No. 18/626,317, filed Apr. 3, 2024, now U.S. Pat. No. 12,097,161, which is a continuation of U.S. patent application Ser. No. 18/526,980, filed Dec. 1, 2023, now U.S. Pat. No. 12,036,174 which is a continuation of U.S. patent application Ser. No. 18/388,489, filed Nov. 9, 2023, now U.S. Pat. No. 11,931,174, which is a continuation of U.S. patent application Ser. No. 18/143,142, filed May 4, 2023, now U.S. Pat. No. 11,864,914, which is a continuation of U.S. patent application Ser. No. 18/092,398, filed Jan. 2, 2023, now U.S. Pat. No. 11,684,313, which is a continuation of U.S. patent application Ser. No. 17/213,117, filed Mar. 25, 2021, now U.S. Pat. No. 11,589,816, which is a continuation of U.S. patent application Ser. No. 16/757,482, filed Apr. 20, 2020, now U.S. Pat. No. 10,959,674, which is a § 371 National Stage

Entry of International Patent Application No. PCT/US18/56814, filed Oct. 22, 2018, claiming the benefit of priority of U.S. Provisional Patent Application No. 62/676,949, filed May 26, 2018, and U.S. Provisional Patent Application No. 62/575,951, filed Oct. 23, 2017, the entireties of which are incorporated by reference into the enclosed application.

TECHNICAL FIELD

(1) Aspects of the present disclosure generally relate to communication devices, methods, and systems.

BACKGROUND

(2) Computer screens have emerged as the most common means for person-to-computer communication. In 2015, for example, it was estimated that the average adult spends roughly 10 hours a day looking at a screen to consume information and/or communicate with others. The human eye was not designed for all this screen time, and numerous health problems have been associated therewith. For example, eyestrain from hours of screen time may cause instances of eye irritation, dryness, fatigue, and/or blurred vision that last for extended periods of time. These problems are increasingly common, and the near constant production of new screen-oriented devices (e.g., the next iPhone®) suggests further increases.

(3) Alternate means for person-to-computer communications may reduce the negative effects of excessive screen time. For example, the human body includes many non-optical nerves that are capable of communicating data to the brain, such as the nerves associated with the skin. Further improvements are required to better leverage these and other communication capabilities of living tissue. Aspects of this disclosure may solve the above reference problems, solve other known problems, and/or overcome other deficiencies in the prior art.

SUMMARY

(4) Numerous aspects are disclosed in this application. One exemplary aspect is a communication device comprising: a body comprising a distal surface compatible with skin; a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; an attachment element configured to maintain the tissue interface against the skin; and a processing unit configured to communicate with nerves associated with the skin by receiving input data from a data source and causing the plurality of energy generators to output an energy signal in the signal direction with one or more energies of the plurality of energies.

(5) The body may be flexible. The body may comprise a plurality of communication bays, and each energy generator may be located in and configured to output the energy signal out of one of the communication bays. The body may comprise an insulating material configured to promote flows of the one or more energies out of each communication bay in the signal direction, and limit flows of the one or more energies between the plurality of communication bays. The attachment element may comprise a plurality of holes aligned with the plurality of communication bays, and each energy generator may be configured to output the plurality of energies through one of the holes. An interior surface of each communication bay or hole may be configured to direct the one or more energies in the signal direction. The interior surface of each communication bay or hole may be configured to focus at least one energy of the one or more energies in the signal direction.

(6) The attachment element may comprise a biocompatible adhesive disposed on the distal surface of the body. The attachment element may comprise an elastic portion configured to maintain the tissue interface against the skin. The elastic portion may expand to receive a circular portion of the skin and contract to maintain the tissue interface against the circular portion of the skin. The body may be removably attached to the attachment element. The input data may comprise a measurement, and the processing unit may be configured to modify at the energy signal based on the measurement. The processing unit may be configured determine a change of the measurement and modify the energy signal based on the change of the measurement.

(7) The plurality of energy generators may be spaced apart on the distal surface of the body in a pattern; each energy generator may be operable to output the one or more energies in the signal direction toward a different point on the pattern; and the energy signal may comprise a plurality of symbols based on the pattern. Each symbol may comprise a plurality of dots, and each dot may correspond with one of the different points on the pattern. The plurality of symbols may comprise at least one alphanumeric symbol. The processing unit may be operable with the plurality of energy generators to scroll the plurality of symbols across the skin at a scroll rate in a communication direction transverse with the signal direction.

(8) The input data may comprise vital signs of a subject, and the plurality of symbols may comprise a symbol associated with each vital sign. The plurality of symbols may comprise a symbol associated with an identity or location of the subject. The processing unit may be configured to: determine a change of the vital signs; and modify the one or both of the plurality of symbols and the scroll rate based on the change of the vital signs. The processing unit may be configured to: output one or more of the plurality of symbols with a first combination of the one or more energies when the change is within a predetermined range; and output the one or more of the plurality of symbols with a second combination of the one or more energies when the change is outside the predetermined range.

(9) The data source may comprise one or more data sources, and the processing unit may be configured to: receive the input data from the one or more data sources; generate a control signal based on the input data; and cause the plurality of energy generators to output the energy signal according to the control signal. The processing unit may be configured to: determine a change in the input data; and modify the control signal based on the change of the input data. The control signal may comprise a scroll rate for the energy signal the processing unit may be configured to: determine the scroll rate based on the input data; and cause the plurality of generators to scroll the energy signal across the skin at the scroll rate.

(10) Each energy generator may comprise a plurality of generator elements and a controller operable with the plurality of generator elements to output the plurality of energies; the control signal may comprise output commands for each controller of each energy generator; and each controller may be configured to receive the control signal, select one of the output commands, and cause one or more of the plurality of generator elements to output the one or more energies based on the selected one of the output commands.

(11) Each energy generator may comprise a plurality of generator elements, and each generator element may be operable to output one of the plurality of energies in the signal direction. The plurality of generator elements may comprise one or more of: an impact generator element; a heat generator element; a shock generator element; and a pressure generator element. Each generator element may be configured to output the one of the plurality of energies toward a similar point or area on the skin. The plurality of generator elements may be arranged coaxially with a communication axis parallel to the signal direction.

(12) The data source may comprise a local sensor that is attached to the body and configured to output a portion of the input data. The data source may comprise at least one remote sensor that is remote from the body and configured to output a portion the input data. The processing unit may be configured to receive the input data from a server in communication with the at least one remote sensor. The at least one remote sensor may comprise a health monitoring device. The device may comprise a power generator attached to the body. The power generator may comprise a photovoltaic cell mounted to a proximal surface of the body. The one or more energies may comprise: a first energy configured to communicate the energy signal; and a second energy configured to modify a penetration depth of the first energy. The first energy may be communicable with a first portion of the nerves, and the second energy may be communicable with a second portion of the nerves.

(13) The body may comprise an impact absorbing material; and the attachment element may

comprise a garment configured to maintain a position of the impact absorbing material relative to a user body. The processing unit may be configured to determine a direction of movement for the user body and output the energy signal based on the direction of movement. The processing unit may be configured to determine a change in the direction of movement and modify the energy signal based on the change in the direction of movement. The attachment element may comprise a shoe, and the distal surface of the body may comprise an interior surface of the shoe. The input data may comprise GPS signals, the processing unit may be configured to determine a direction of movement for a user body based on the GPS signals.

(14) The attachment element may comprise a grip, and the body may comprise an exterior surface of the grip. The grip may be integral with a weapon comprising a sight, the input data may comprise data associated with an orientation of the sight, and the energy signal may be configured to communicate a status of the weapon based on the orientation of the sight. The input data may comprise data associated with an alignment of the sight with a target, and the processing unit may be configured to output the energy signal with a first combination of the one or more energies when the target is not aligned with the sight and a second combination of the one or more energies when the target is aligned with the sight.

(15) The device may be implantable. The attachment element may comprise a bone plate engageable with a bone to orient the tissue interface toward an underside of the skin. The attachment element may comprise a tissue in-growth structure interactable with living tissue to maintain an orientation of the tissue interface toward an underside of the skin.

(16) The plurality of energy generators may be arranged in bands; the attachment element may be configured to maintain each band against the skin; the input data may comprise input data for each band; and the processing unit may be configured to communicate with nerves associated with the skin by causing the plurality of energy generators in each band to output a different energy signal based on the input data for each band. The body may extend along a longitudinal axis; and the bands may be spaced apart along the longitudinal axis. The body may be configured to wrap around a limb so that the longitudinal axis of body is aligned with a longitudinal axis of the limb, and the bands wrap around the limb about the longitudinal axis. The processing unit may be operable with the plurality of energy generators to scroll each different energy signal in each band in a communication direction transverse with the signal direction. The input data for each band may comprise different vital signs, and each different energy signal may be based on one of the different vital signs.

(17) The signal direction may comprise a first signal direction and the device may comprise an optical interface on a proximal surface of the body; the optical interface may comprise at least one display element operable to output at least one color toward eyes in a second signal direction opposite of the first signal direction; and the processing unit may be operable with the tissue interface and the optical interface to communicate simultaneously with nerves associated the skin and the eyes by outputting the energy signal with the one or more energies of the plurality of energies in the first signal direction and outputting an optical signal with the at least one color in the second signal direction.

(18) The body may extend along a longitudinal axis, and the first and second signal directions may be transverse with the longitudinal axis. The energy signal and the optical signal may be scrolled together along or about the longitudinal axis. The processing unit may be configured to: receive the input data from the data source; generate a control signal based on the input data; cause the plurality of energy generators to output the energy signal according to the control signal; and cause the at least one display element to simultaneously output the optical signal according to the control signal. The energy signal may correspond with the optical signal.

(19) The input data may comprise vital signs of a subject, the energy signal may comprise a plurality of symbols associated with the vital signs, and the optical signal may comprise the plurality of symbols. The processing unit may be configured to: determine a change of the vital

signs; and modify the plurality of symbols based on the change of the vital signs. The device may comprise a motion sensor attached to the body, and the processing unit may be configured to selectively output the optical signal in response to the motion sensor.

(20) Another exemplary aspect may comprise a system. For example, the system may comprise: a plurality of any communication devices described herein; and at least one processor that is in communication with the plurality of communication devices and configured to: generate a corrective motion signal based on position data for the plurality of communication devices; and cause each communication device to output its energy signal based on the corrective motion signal.

(21) The system may comprise at least one position sensor configured to determine the position data and output the position data to the at least one processor. The position data may comprise an actual location of each communication device; and the at least one processor may be configured to generate the corrective motion signal based on the actual location of each communication device and a target location of each communication device. The position data may comprise an actual spatial configuration of the plurality of communication devices; and the at least one processor may be configured to generate the corrective motion signal based on the actual spatial configuration and a target spatial arrangement for the plurality of communication devices. The at least one processor may be configured to: determine a movement direction for each communication device based on the actual and target spatial arrangements; and cause each communication device to scroll its energy signal across the skin in the movement direction.

(22) Each communication device may be mounted to a different part of a user body; and the target spatial arrangement may comprise a physical position of the user body defined by relative positions of each different part of the user body. The physical position of the user body may comprise a pose or a stance. The at least one processor may be configured to guide the user body through a series of different positions by determining the movement direction at intervals and modifying the energy signal for each interval.

(23) Another exemplary aspect may comprise another communication device. In keeping with above, the device may comprise: a body comprising a distal surface compatible with skin; a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and a processing unit configured to communicate with nerves associated with the skin by receiving input data, and causing the plurality of energy generators to output one or more energies of the plurality of energies in the signal direction.

(24) The body may be flexible. The device may further comprise an attachment element configured to maintain the tissue interface in a position on or adjacent the skin. The distal surface of the body may comprise a biocompatible adhesive that is adherable to the skin. The body may comprise a plurality of communication bays, and each energy generator may be located in one of the communication bays. The attachment element may comprise a plurality of holes aligned with the plurality of communication bays, and each energy generator may be configured to output the plurality of energies through one of the holes. An interior surface of each communication bay or hole may be configured to direct the one or more energies in the signal direction. The interior surface may be configured to focus the at least one energy of one or more energies. The attachment element may comprise an elastic band. The body may be removably attached to the attachment element.

(25) Another exemplary aspect may comprise another communication device. The communication device may comprise: a body comprising a distal surface compatible with skin; a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; an attachment element configured to maintain the tissue interface against the skin; and a processing unit configured to communicate an energy signal to nerves associated with the skin by receiving input data, and causing the plurality of energy generators to output one or more energies of the

plurality of energies in the signal direction.

(26) The plurality of energy generators may be spaced apart in a grid pattern, and each energy actuator may be operable to output the plurality of energies towards a different point on the grid pattern. The energy signal may comprise a plurality of symbols, each symbol may comprise a plurality of dots, and each dot may correspond with one of the different points on the grid pattern. The plurality of dots in each symbol may be arranged in a dot pattern within the grid pattern. The processing unit may be operable with the plurality of energy generators to scroll the plurality of symbols across the skin in a communication direction transverse with the signal direction. For example, the processing unit may be operable with the plurality of energy generators to output and scroll each symbol using a different combination of the one or more energies of the plurality of energies.

(27) The input data may comprise a measurement, and the processing unit may be configured to communicate the energy signal by selecting the one or more energies of the plurality of energies based on the measurement. For example, the processing unit may be configured to communicate the energy signal by determining a change of the measurement and modifying the one or more energies of the plurality of energies based on the change of the measurement.

(28) Another exemplary aspect may comprise another communication device. For example, the communication device may comprise: a body comprising a distal surface compatible with skin; a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; an attachment element configured to maintain the tissue interface on or adjacent the skin; and a processing unit configured to communicate an energy signal to nerves associated with the skin by: (i) receiving input data; (ii) selecting one or more energies of the plurality of energies based on the input data; and (iii) causing the plurality of energy generators to output the one or more energies in the signal direction.

(29) The processing unit may be further configured to communicate the energy signal by: (iii) determining a change in the input data; and (iv) modifying the one or more energies based on the change. The processing unit may be further configured to communicate the energy signal by: (v) selecting a scroll rate based on the input data; and (vi) causing the plurality of generators to scroll the one or more energies across the skin at the scroll rate. The energy signal may comprise a plurality of symbols scrolled across the skin in a communication direction transverse with the signal direction. At least one symbol of the plurality of symbols may be an alphanumeric symbol.

(30) Another exemplary aspect may comprise a communication method. The method may comprise: receiving, with a processing unit, input data for a communication device comprising a tissue interface maintainable on or adjacent skin, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and operating, with the processing unit, the plurality of energy generators to communicate with nerves associated with the skin by outputting one or more energies of the plurality of energies in the signal direction based on the input data.

(31) The receiving step may comprise receiving the input data from one or more data sources. For example, the one or more data sources may comprise at least one of patient monitoring device, a remote server, and a sensor. The receiving step may comprise receiving the input data from the one or more data sources at regular intervals, and the operating step may comprise outputting the one or more energies based on the input data received during each regular interval. The input data may comprise a control signal, and the operating step may comprise outputting the one or more energies based on the control signal.

(32) The method may further comprise generating, with the processing unit, a control signal based on the input data, wherein the operating step comprises outputting the one or more energies based on the control signal. Generating the control signal may comprise associating the input data with a plurality of symbols, and the operating step may comprise communicating the plurality of

symbols to the skin with the one or more energies. For example, the input data may comprise vital signs of a patient, and each symbol may be associated with one or more of the vital signs. The one or more energies may comprise a first combination of the plurality of energies followed by a second combination of the plurality of energies. The one or more energies also may comprise a first energy communicable with a first portion of the nerves, and a second energy communication with a second portion of the nerves.

(33) Another exemplary aspect may comprise another communication method. The method may comprise: receiving, with a processing unit, input data for a communication device comprising a tissue interface maintainable on or adjacent skin, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and operating, with the processing unit, the plurality of energy generators to communicate an energy signal to nerves associated with the skin by outputting one or more energies of the plurality of energies in the signal direction based on the input data.

(34) The operating step may comprise outputting different combinations of the one or more energies, and each different combination may communicate a different portion of the energy signal. The energy signal may comprise one or more symbols, and the operating step may comprise outputting the one or more energies to communicate the one or more symbols. The operating step may comprise scrolling the one or more symbols across the skin in a communication direction transverse with the signal direction. The one or more symbols may comprise an alphanumeric symbol.

(35) The operating step may comprise: outputting a first combination of the one or more energies to communicate a first symbol of the one or more symbols, and outputting a second combination of the one or more energies to communicate a second symbol of the one or more symbols. The operating step may comprise: outputting a first combination of the one or more energies to communicate the energy signal, and outputting a second combination of the one or more energies to communicate a characteristic of the energy signal. The input data may comprise a measurement, and the operating step may comprise outputting the one or more energies based on the measurement. For example, the operating step may comprise modifying the one or more energies based on a change of the measurement.

(36) Another exemplary aspect may comprise another communication method. For example, the method may comprise: receiving, with a processing unit, input data for a communication device comprising a tissue interface maintainable on or adjacent skin, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; generating, with the processing unit, a control signal based on the input data; and operating, with the processing unit, the plurality of energy generators to communicate with to nerves associated with the skin by outputting one or more energies of the plurality of energies in the signal direction based on the control signal.

(37) Another exemplary aspect may comprise another communication device. The device may comprise: a body comprising a distal surface compatible with skin; a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators arranged in bands, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and a processing unit configured to communicate with nerves associated with the skin by receiving input data, and causing the plurality of energy generators in each band to output one or more energies of the plurality of energies in the signal direction.

(38) The body may be flexible. The device may further comprise an attachment element configured to maintain the tissue interface in a position on or adjacent the skin. For example, the attachment element may comprise a distal surface adherable to the skin. The attachment element may be proximal of the tissue interface and configured to maintain the bands against the skin. The attachment element may be configured to maintain the bands against the skin by applying a tensile force to the body.

(39) Another exemplary aspect may comprise another communication device. The device may comprise: a body extending along a longitudinal axis, and comprising a distal surface compatible with skin; a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators arranged in bands spaced apart along the longitudinal axis, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; an attachment element configured to maintain the bands of the tissue interface against the skin; and a processing unit configured to communicate energy signals to nerves associated with the skin by receiving input data, and causing the plurality of energy generators to output an energy signal in each band with one or more energies of the plurality of energies.

(40) The body may be configured to wrap around a limb so that the longitudinal axis of body is aligned with a longitudinal axis of the limb, and the bands wrap around the limb about the longitudinal axis. The processing unit may be configured to move the energy signal in each band so as to scroll the energy signal around the limb. The one or more energies may comprise: a first energy configured to communicate one or more symbols; and a second energy configured to modify the one or more symbols.

(41) Another exemplary aspect may comprise another communication method. The method may comprise: receiving, with a processing unit, input data for a communication device comprising a tissue interface maintainable on or adjacent skin, the tissue interface comprising a plurality of energy generators arranged in bands, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and causing, with the processing unit, the plurality of energy generators in each band to communicate with nerves associated with the skin by outputting one or more energies of the plurality of energies in response to the input data.

(42) The receiving step may comprise receiving the input data from one or more data sources. For example, the receiving step may comprise: receiving input data comprise a plurality of measurements; and causing the plurality of energy generators in each band to output the one or more energies based on one measurement of the plurality of measurements. The method may further comprise outputting a first combination of the one or more energies when the one measurement is inside of an acceptable range; and outputting a second combination of the one or more energies when the one measurement is outside of the acceptable range.

(43) The receiving step may comprise receiving input data comprising a plurality of vital signs; and the causing step may comprise causing the plurality of energy generators in each band to output the one or more energies based on one vital sign of the plurality of vital signs. The input data may comprise a control signal for each band, and the operating step may comprise outputting the one or more energies based on the control signal for each band. The method may further comprise generating, with the processing unit, a control signal for each band based on the input data, wherein the operating step may comprise outputting the one or more energies based on the control signal for each band.

(44) Another exemplary aspect may comprise another communication method. The method may comprise: receiving, with a processing unit, input data for a communication device comprising a tissue interface maintainable on or adjacent skin, the tissue interface comprising a plurality of energy generators arranged in bands, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and causing, with the processing unit, the plurality of energy generators to communicate energy signals to nerves associated with the skin by outputting an energy signal in each band with one or more energies of the plurality of energies. The energy signal may comprise one or more symbols based on the input data, and the operating step may comprise outputting the one or more symbols to the skin with one or more energies. The operating step may comprise scrolling the one or more symbols across the skin in a communication direction transverse with the signal direction.

(45) Another exemplary aspect may comprise a communication system. The system may comprise: (A) a plurality of communication devices, each communication device comprising: a body

comprising a distal surface compatible with skin; and a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and (B) a processing unit in communication with at least one of the plurality of communication devices and configured to: generate, with one or more processors, a corrective motion signal based on position data for the plurality of communication devices; and operate, with the one or more processors, the plurality of energy generators of each communication device to output one or more energies of the plurality of energies in the signal direction based on the corrective motion signal.

(46) The system may further comprise at least one position sensor configured to determine the position data and output the position data to the processing unit. The position data may comprise an actual location of each device of the plurality of communication devices; and the processing unit may be configured to generate, with the one or more processors, the corrective motion signal based on the actual locations and a target location for each device of the plurality of communication devices.

(47) The position data may comprise an actual spatial arrangement of the plurality of communication devices; and the processing unit may be configured to generate, with the one or more processors, the corrective motion signal based on the actual spatial arrangement and a target spatial arrangement for the plurality of communication devices. The processing unit may be configured to: determine, with the one or more processors, a movement direction for each communication device based on the actual and target spatial arrangements; and operate, with the one or more processors, the plurality of energy generators of each communication device to output the one or more energies toward the skin in the signal direction and move the one or more energies across the skin the movement direction.

(48) Another exemplary aspect may comprise another communication method. The method may comprise: generating, with one or more processors, a corrective motion signal for a plurality of communication devices based on position data, each communication device comprising a tissue interface with a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; and operating, with the one or more processors, the plurality of energy generators of each communication device to output one or more energies of the plurality of energies in the signal direction based on the corrective motion signal. The method may comprise receiving the position data from the plurality of communication devices or a remote position sensor.

(49) Another exemplary aspect may comprise another communication method. The method may comprise: receiving, with one or more processors, position data for a plurality of communication devices mountable on or adjacent skin, each device comprising a tissue interface with a plurality of energy generators, each energy generator being operable to output a plurality of energies in a signal direction toward the skin; receiving or generating, with the one or more processors, a corrective motion signal for the plurality of communication devices based on position data for each communication device; and operating, with the one or more processors, the plurality of energy generators of each communication device to output one or more energies of the plurality of energies in the signal direction based on the corrective motion signal.

(50) At least one of the communication devices may comprise a position sensor, and the method may comprise receiving, with the one or more processors, the position data from the position sensor. The method may comprise: determining, with the one or more processors, an actual spatial arrangement of the plurality of communication devices based on the position data; and identifying, with the one or more processors, a target spatial arrangement for the plurality of communication devices, wherein the generating step comprises generating, with the one or more processors, the corrective motion signal based on the actual spatial arrangement and the target spatial arrangement. The method may comprise: determining, with the one or more processors, a movement direction for each communication device based on actual and target spatial arrangements; and operating,

with the one or more processors, the plurality of energy generators of each communication device to output one or more energies toward the skin in a shape associated with the movement direction for each communication device.

(51) The method may comprise operating, with the one or more processors, the plurality of energy generators of each communication device to move the shape across the skin in the movement direction. Each communication device may be mounted to a different portion of a body; and the target spatial arrangement may comprise a physical position of the body defined by the relative positions of each different portion of the body. The physical position of the body may comprise at least one of a stretching position, a lifting position, a pose, or a stance.

(52) The target spatial arrangement may comprise a series of target spatial arrangements, and the method may comprise: selecting arrangements from the series of target spatial arrangements; and repeating the determining, generating, and operating steps for each selected arrangement. The selecting step may be performed at predetermined intervals so as to coordinate relative movements between each selected arrangement. The series of target spatial arrangements may comprise one or more stretches, yoga poses, or defensive postures.

(53) Another exemplary aspect may comprise another communication device. The device may comprise: a body comprising a proximal surface compatible with eyes, and a distal surface compatible with skin; a tissue interface on the distal surface, the tissue interface comprising a plurality of energy generators, each energy generator comprising a tissue interface operable to output a plurality of energies in a first signal direction toward the skin; an optical interface on the proximal surface, the optical interface comprising at least one display element operable to output at least one color in a second signal direction toward the eyes; and a processing unit operable with the tissue interface and the optical interface to communicate simultaneously with nerves associated with skin and eyes by outputting one or more energies of the plurality of energies in the first signal direction and at least one color in the second signal direction.

(54) The body may extend along a longitudinal axis, and the first signal direction may be transverse with the longitudinal axis. The second signal direction may be transverse with the longitudinal axis. The first and second signal directions may extend oppositely along a signal axis transverse with the longitudinal axis. The body may be conformable with a curved shape. The body comprises a flexible body configured to wrap around a limb so that the longitudinal axis has a circular shape.

(55) The processing unit may receive input data from one or more sources, the one or more energies may be output as an energy signal based on the input data, and the one or more colors may be simultaneously output as an optical signal based on the input data. The energy signal may correspond with the optical signal. The outputs may be flashed or scrolled together. For example, the optical signal and the energy signal may be scrolled together along the longitudinal axis. The input data may comprise a vital sign of the patient, the energy signal may communicate the vital sign to the skin, and the optical signal may simultaneously communicate the vital sign to the eyes. The processing unit may be configured to determine a change of the vital sign over time and simultaneously modify one or both of the optical signal and the energy signal based on the change. The input data may comprise alphanumeric symbols, the optical signal may communicate the alphanumeric symbols to the eyes, and the energy signal may simultaneously communicate the symbols to skin.

(56) Another exemplary aspect may comprise another communication device. The device may comprise: a body extending along a longitudinal axis, the body comprising a proximal surface compatible with eyes and a distal surface compatible with skin; a tissue interface on the distal surface of the body, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a first signal direction toward the skin; an optical interface on the proximal surface, the optical interface comprising at least one display element operable to output at least one color in a second signal direction toward the eyes; a sensor on the body; and a processing unit configured to communicate simultaneously with nerves

associated with the eyes and the skin by: receiving input data from the sensor or a remote data source, causing the plurality of energy generators to output one or more energies of the plurality of energies in the first signal direction as an energy signal, and causing the display element to output the at least one color in the second signal direction as an optical signal.

(57) The first and second communication signals may be scrolled together along the longitudinal axis. The energy signal may be output continuously. The optical signal may be output in response to a movement detected by the sensor. The movement may comprise aligning the optical interface with the eyes.

(58) Another exemplary aspect may comprise another communication device. The device may comprise: a body comprising a proximal surface compatible with eyes and a distal surface compatible with skin; a tissue interface on the distal surface of the body, the tissue interface comprising a plurality of energy generators, each energy generator being operable to output a plurality of energies in a first signal direction toward the skin; at least one sensor; an optical interface on the proximal surface, the optical interface comprising at least one display element operable to output at least one color in a second signal direction toward the eyes; a processing unit configured to communicate simultaneously with nerves associated with the eyes and the skin by: receiving vital sign data from the at least one sensor, causing the plurality of energy generators to output one or more energies of the plurality of energies in the first signal direction as an energy signal, and causing the at least one display element to output the at least one color in the second signal direction as an optical signal.

(59) The energy signal and the optical signal may be scrolled across the body in a communication direction transverse with the longitudinal axis. The at least one display element may comprise a touchscreen, and the energy signal may be moveable together with optical signal along or around the longitudinal axis by operation of the touchscreen. The first direction may be transverse with the second direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are incorporated in and constitute a part of this specification. These drawings illustrate exemplary aspects of the present disclosure that, together with the written descriptions provided herein, serve to explain the principles of this disclosure.

(2) FIG. 1A depicts an exemplary energy signal output onto a living tissue;

(3) FIG. 1B depicts an exemplary communication device configured to output the energy signal of FIG. 1A;

(4) FIG. 2A depicts a top-down view of the FIG. 1B device;

(5) FIG. 2B depicts a bottom-up view of the FIG. 1B device;

(6) FIG. 2C depicts a cross-section view of the FIG. 1B device taking along section line A-A of FIG. 2A;

(7) FIG. 3A depicts a cross-section of an exemplary energy generator;

(8) FIG. 3B depicts a bottom-up view of the FIG. 3A generator;

(9) FIG. 4A depicts an impact energy output with the FIG. 3A generator;

(10) FIG. 4B depicts a heat energy output with the FIG. 3A generator;

(11) FIG. 4C depicts an electrical energy output with the FIG. 3A generator;

(12) FIG. 4D depicts a pressure energy output with the FIG. 3A generator;

(13) FIG. 5 depicts an exemplary processing unit;

(14) FIG. 6A depicts another exemplary communication device;

(15) FIG. 6B depicts another exemplary communication device;

(16) FIG. 6C depicts another exemplary communication device;

- (17) FIG. 6D depicts another exemplary communication device;
- (18) FIG. 7A depicts another exemplary communication device;
- (19) FIG. 7B depicts another exemplary communication device;
- (20) FIG. 7C depicts another exemplary communication device;
- (21) FIG. 7D depicts another exemplary communication device;
- (22) FIG. 8A depicts another exemplary communication device;
- (23) FIG. 8B depicts a cross-section view of the FIG. 8A device;
- (24) FIG. 9 depicts an exemplary method;
- (25) FIG. 10 depicts another exemplary method;
- (26) FIG. 11 depicts an exemplary communication system;
- (27) FIG. 12 depicts another exemplary method;
- (28) FIG. 13A depicts another exemplary communication device;
- (29) FIG. 13B depicts another view of the device of FIG. 13A;
- (30) FIG. 14A depicts another view of the device of FIG. 13A; and
- (31) FIG. 14B depicts another view of the device of FIG. 13A.

DETAILED DESCRIPTION

(32) Aspects of the present disclosure are now described with reference to exemplary communication devices, methods, and systems. Particular aspects reference a healthcare setting, wherein the described devices, methods, and systems may allow a single caregiver to monitor vital signals for a plurality of patients without using a screen, or at least with a reduced amount of screen time. Any references to a particular setting, such as healthcare; a particular user, such as a caregiver; a particular data, such as vital signals; or particular amount of screen time, are provided for convenience and not intended to limit the present disclosure unless claimed. Accordingly, the aspects disclosed herein may be utilized for any analogous communication device, method, or system—healthcare-related or otherwise.

(33) The terms “proximal” and “distal,” and their respective initials “P” and “D,” may be used to describe relative components and features. Proximal may refer to a position closer to a hand of user, whereas distal may refer to a position further away from said hand. With respect to a hand adjacent a living tissue, for example, proximal may refer to a position away from the tissue, whereas distal may refer to a position toward said tissue. As a further example, with respect to energy directed toward the living tissue, proximal may refer to energy directed away from the tissue and distal may refer to energy directed toward the tissue. Appending the initials P or D to a number may signify its proximal or distal location or direction. Unless claimed, these directional terms are provided for convenience and not intended to limit this disclosure.

(34) Aspects of this disclosure may be described with reference to one or more axes. For example, an element may extend along an axis, be moved along said axis in first or second direction, and/or be rotated about said axis in a first or second direction. One axis may intersect another axis, resulting in a transverse and/or perpendicular relationship therebetween. For example, two or three perpendicular axes may intersect at an origin point to define a Cartesian coordinate system. The directional terms proximal and distal may be used with reference to any axis. One axis may be a longitudinal axis extending along a length of an element, such as a central longitudinal axis extending along the length and through a centroid of the element.

(35) Terms such as “may,” “can,” and like variation, are intended to describe optional aspects of the present disclosure, any of which may be covered by the claims set forth below. Terms such as “comprises,” “comprising,” or like variation, are intended to describe a non-exclusive inclusion, such that a device, method, or system comprising a list of elements does not include only those elements, but may include other elements not expressly listed or inherent thereto. The term “and/or” indicates a potential combination, such that a first and/or second element may likewise be described as a first element, a second element, or a combination of the first and second elements. These potential combinations are provided as examples. Numerous other combinations are inherent

to this disclosure. Unless stated otherwise, the term “exemplary” is used in the sense of “example” rather than “ideal.”

(36) Aspects of this disclosure are directed to devices, methods, and systems for communicating with the brain through nerves associated with a living tissue. Some aspects are described with reference to an energy signal including one or more energies output to communicate symbols to the living tissue. The symbols may be used to communicate data, and the one or more energies may be used to communicate aspects of the data. The living tissue may be a portion of skin, as shown in FIGS. 1A-8D. In a healthcare setting, the energy signal may be output towards the skin of a caregiver to communicate symbols associated with a status of a patient. For example, an intensity of the one or more energies may escalate responsive to a measure of the status, providing a non-visual alert to the caregiver if the measure changes.

(37) Exemplary energies and energy signals are now described with reference to FIG. 1A, which depicts an exemplary energy signal **90** including a plurality of symbols **92** output onto a communication area **4** of a skin **2** with one or more energies **32**. For illustrative purposes, the symbols **92** of FIG. 1 are shown from a proximal-to-distal direction, as they would be output to skin **2** by an energy transceiver. Each energy **32** may be configured to communicate aspects of the data to the brain through nerves associated with skin **2**, such as nerves located distal of communication area **4**. For example, the one or more energies **32** shown in FIG. 1A may be recognizable by touch receptors, such as the Meissner's corpuscle; temperature receptors, such the Ruffini corpuscle and Krause corpuscle; electrical receptors, such as the muscles and pain receptors located in the dermis layer; pressure receptors such as the Pacinian corpuscle; and/or other cutaneous or subcutaneous nerves that innervate the skin or other living tissue.

(38) Each symbol **92** may be associated with different data. For example, in the healthcare setting, each symbol **92** may be associated with a vital sign of the patient, such as body temperature, pulse rate, respiration rate, and/or blood pressure. As shown in FIG. 1A, the plurality of symbols **92** may include a first symbol **92A**, a second symbol **92B**, and a third symbol **92C**. In keeping with the previous example, first symbol **92A** may be associated with temperature and pulse rate, second symbol **92B** may be associated with respiration rate, and third symbol **92C** may be associated with blood pressure. Any number of symbols **92** may be provided and/or associated with a measurable or non-measurable characteristic of the patient.

(39) Symbols **92A**, **92B**, and **92C** are shown as pip patterns of dots in FIG. 1A, wherein each dot is a shaded area. Each dot may represent an output of the one or more energies **32**. Aspects of energies **32** and/or each symbol **92A**, **92B**, and **92C** may increase the complexity of energy signal **90**, and thus the amount of data transmitted therewith. As shown in FIG. 1A, symbols **92A**, **92B**, and **92C** may be scrolled across communication area **4** by outputting energies **32** toward the skin in the pip patterns; and moving the patterns across the skin in a communication direction CD. In FIG. 1A, first symbol **92A** is a pip five dot pattern; second symbol **92B** is a pip six dot pattern; and a third symbol **92C** is a pip three dot pattern that has been truncated by an end of communication area **4** due to the scrolling. Symbols **92** may be flashed and scrolled. For example, the five dots of first symbol **92A** in FIG. 1A may be output to communicate a temperature range of the patient (e.g., a normal range), and flashed on-and-off to communicate the pulse rate of the patient.

(40) An exemplary energy transceiver **10** is depicted in FIG. 1B as being configured output energy signal **90** to communication area **4** of skin **2**. As shown, energy transceiver **10** may be attached to a portion of skin **2**, including any portion located on a limb, such as the underside of a human wrist shown in FIG. 1B for example. Communication area **4** may be sized approximate to a perimeter of transceiver **10**. In this configuration, transceiver **10** may be configured to communicate energy signal **90** to skin **2** by outputting the one or more energies **32** toward communication area **4** in a signal direction oriented toward skin **2**. As shown in FIG. 1A, the energies **32** may be output individually and/or in combination to communicate aspects of any of symbols **92A**, **92B**, and **92C** to skin **2**.

(41) Additional aspects of exemplary energy transceiver **10** are now described with reference to FIGS. 2A-C. As shown, transceiver **10** may comprise: a body **20**; a tissue interface **30**; a processing unit **60**; and an attachment element **70**. With these elements, and the variations described herein, energy transceiver **10** may be configured to communicate energy signal **90** to nerves associated with skin **2** by outputting the one or more energies **32** towards skin **2** with tissue interface **30**.

(42) As shown in FIGS. 2A-C, body **20** may contain the elements of energy transceiver **10**. For example, body **20** of FIGS. 2A-C has a length extending along a longitudinal axis X-X, a width extending along a lateral axis Y-Y, and a thickness extending along a proximal-distal axis Z-Z. The length, width, and/or thickness of body **20** may be compatible with skin **2**. For example, body **20** may be composed of a flexible biocompatible base material, such as a polymeric material, so that the length and width of body **20** are conformable against a curvature of skin **2**.

(43) Body **20** may include any shape and be conformable with any curvature. For example, body **20** may be conformable with a cylindrical shape of a human forearm (e.g., FIG. 1B), a semi-spherical shape a human forehead (e.g., FIG. 6B), or an irregular curved shape of a human foot (e.g., FIG. 7A). A plurality of bodies **20** may be joined together to accommodate some curvatures. For example, side surfaces of body **20** of FIGS. 2A-C may be removable engageable with side surfaces of additional bodies **20** to create a joined layer conformable with the curvature.

(44) The base material of body **20** may have insulating and/or energy-directing properties. For example, the base material may include compositions and/or coatings that promote energy flows along proximal-distal axis Z-Z, and limit energy flows along axes X-X and/or Y-Y. Body **20** may be manufactured from the base material using any known process. For example, body **20** may be molded or 3D printed from a base material that is biocompatible, dielectric, impact resistance, sound absorbing, and/or thermally resistant, such as polyether ether ketone (PEEK) and like polymeric materials. Additional materials and/or coatings may be included with the base material and/or applied to body **20** to further promote biocompatibility.

(45) As shown in FIGS. 2A-C, body **20** may define a proximal surface **22** (FIG. 2A) opposite of a distal surface **24** (FIG. 2B) along proximal-distal axis Z-Z (FIG. 2C). In FIGS. 2A and 2C, for example, proximal surface **22** includes a processor compartment **23** configured to receive processing unit **60**. As shown, and described further below, processing unit **60** may be removable engageable (e.g., snap-fit into) with processor compartment **23**. Body **20** may include and/or be compatible with additional mechanisms for securing and/or releasing the snap-fit, such as a retaining screw and/or a lever.

(46) Body **20** of FIGS. 2A-C includes a plurality of communication bays **25**. As shown, each communication bay **25** may be spaced apart from the next on distal surface **24** in a grid pattern. The spacing may be uniform or non-uniform. In FIGS. 2B and 2C, the bays **25** are spaced apart uniformly for communication with the skin **2** of FIG. 1B, which has a fairly planar surface area. Non-uniform spacing may be used to accommodate a curvature of skin **2**. As shown in FIG. 2C, each communication bay **25** may extend proximally into body **20** through distal surface **24** along a communication axis z-z that is parallel with the proximal-distal axis Z-Z of transceiver **10**. In FIG. 2C, a conduit **26** extends proximally from each bay **25**, through an interior portion of body **20**, and into processor compartment **23**, placing the plurality of bays **25** in communication with compartment **23**.

(47) Aspects of tissue interface **30** are now described with reference to FIGS. 2B and 2C. As shown, tissue interface **30** may include a plurality of energy generators **31**, and each generator **31** may be located in one of communication bays **25**. Each generator **31** may be operable with processing unit **60** to output energies **32** individually and/or in combination. In FIGS. 2B and 2C, for example, the one or more energies **32** are being output from the shaded generators **31** to communicate energy signal **90** of FIG. 1A. As shown in FIG. 2C, one or more conductors **27** may extend through each conduit **26** to connect processing unit **60** to each energy generator **31**, allowing control signals to be transmitted between processing unit **60** and the plurality of energy generators

31 along one or more pathways.

(48) As shown in FIG. 2C, the one or more conductors **27** may include any number of electrical wires and/or optical fibers configured to transmit the control signals. For example, the conductors **27** may comprise a plurality of electrical conductors interconnecting the plurality of generators **31** with processing unit **60**, and allowing electricity-based control signals, energies, and communications to be transmitted between unit **60** and generators **31**. In addition or alternatively, the conductors **27** may comprise a plurality of optical fibers interconnecting the plurality of generators with processing unit **60**, and allowing light-based control signals, energies, and communications to be transmitted between unit **60** and generators **31**. For example, each conductor **27** may comprise a twisted pair including at least one electrical conductor and at least one optical fiber. A flexible energy-insulating medium, such as an epoxy, may be used to seal conductors **27** in conduits **26**.

(49) A cross-section of an exemplary energy generator **31** is depicted in FIG. 3A. As shown, each generator **31** may include: a housing **33**; a controller **34**; and a plurality of generator elements, such as: an impact generator element **36**; a heat generator element **42**; a shock generator element **48**; and a pressure generator element **52**. Examples of each generator element are now described.

(50) Similar to body **20**, housing **33** may include an insulating material that surrounds portions of each generator **31** and/or defines mounting surfaces for generator elements **36**, **42**, **48**, and/or **52**. For example, housing **33** may be made of the same base material as body **20** or a compatible material; and/or formed together with body **20** by a molding, printing, or like process. As described below, portions of each generator element **36**, **42**, **48**, and/or **52** may extend distally from housing **33** to contact skin **2**. Housing **33** of FIG. 3A includes an attachment feature **33A** configured to secure each generator **31** in one of the communication bays **25**. For example, attachment feature **33A** may include a set of threads on housing **33** that are engageable with an interior surface of bays **25**. Other types of chemical or mechanical attachment may be used, including biocompatible adhesives, snap-fit connections, and the like.

(51) Exemplary generator elements **36**, **42**, **48**, and **52** may be arranged to output their respective energies **32** in approximately the same direction. As shown in FIGS. 3A and 3B, each generator element **36**, **42**, **48**, and **52** may be arranged coaxially with communication axis z-z so that each energy **32** may be output toward skin **2** in signal direction SD. Because of this coaxial configuration, each energy **32** may be output toward approximately the same point or area on skin **2**, making the energies **32** interchangeable. For example, any of the dots included in energy signal **90** of FIG. 1A may be interchangeably communicated to approximately the same point on skin **2** with any of the energies **32**.

(52) As shown in FIG. 3A, controller **34** may be configured receive a control signal **82** from processing unit **60**, and activate generator elements **36**, **42**, **48**, and **52** according to signal **82**. The one or more conductors **27** may transmit the control signal **82** to generator elements **36**, **42**, **48**, and **52** from processing unit **60** and/or direct electricity to generator elements **36**, **42**, **48**, and **52** from a power source **66** of processing unit **60** (e.g., FIG. 5). Energy transceiver **10** may be an all-electrical device, wherein control signal **82** is an electrical signal and first and the conductors **27** are electrical wires. For varied response times, and energy requirements, transceiver **10** also may be an electro-optical device, wherein control signal **82** includes an optical signal, and at least one of the conductors **27** includes an optical fiber. For example, controller **34** may receive control signal **82** from processing unit **60** with a first one of conductors **27** (e.g., a first electrical and/or optical conductor), and direct electricity to one or more of the generator elements **36**, **42**, **48**, and **52** with a second one of conductors **27** (e.g., a second electrical conductor) according to signal **82**.

(53) Additional aspects of generator elements **36**, **42**, **48**, and **52** are now described with reference to FIGS. 4A-D. As shown in FIG. 4A, for example, impact generator element **36** may be configured to communicate an impact energy **32A** to the brain through nerves associated with skin **2**. For example, impact generator element **36** may be a mechanical actuator that converts electricity

from power source **66** into a mechanical movement recognizable by touch receptors of skin **2**, such as Meissner's corpuscle. As shown, generator element **36** may include a drive mechanism **37**, a piston **38**, a tissue contact **39**, and a guide tube **40**. Drive mechanism **37** may include a motor assembly that is attached to controller **34** and conductively engaged therewith. In this configuration, controller **34** may direct electricity to drive mechanism **37**, causing the motor assembly to move piston **38** distally along communication axis z-z, outputting impact energy **32A** in signal direction SD. Different force transfer components also may be used to apply energy **32A**, including levers and like actuators.

(54) As shown, drive mechanism **37** may be configured to move piston **38** between a retracted position, wherein tissue contact **39** is contained housing **33** (e.g., FIG. 3A); and an extended position, wherein at least a portion of contact **39** is distal of housing **33** (e.g., FIG. 4A). Accordingly, impact energy **32A** may be output in signal direction SD as a physical movement of skin **2** caused by moving tissue contact **39** distally. Aspects of impact energy **32A** may be modified. For example, outer tube **40** may be attached to housing **33** and include interior surfaces configured to modify the timing of energy **32A** by guiding the proximal-distal movements of tissue contact **39** (e.g., by rotating or stabilizing contact **39**). A resilient element may be added between drive mechanism **37** and contact **39** to dampen such movements.

(55) Heat generator element **42** may be configured to communicate a heat energy **32B** to the brain through nerves associated with skin **2**. As shown in FIG. 4B, generator element **42** may include an electrical resistor that converts electricity from power source **66** into an amount of heat recognizable by temperature receptors of skin **2**, such the Ruffini corpuscle. For example, heat generator element **42** may include an electrical resistor **43**, a heat reflecting groove **44**, a conductor **45**, and an insulating material **46**. Groove **44** may include a metal plate attached to an exterior surface of outer tube **40** of generator element **36**. Resistor **43** may include an electrical wire or coil attached to groove **44**. Conductor **45** may include an electrical wire extend between controller **34** and resistor **43**, and material **46** may including an epoxy surrounding conductor **45**.

(56) As shown in FIG. 3B, electrical resistor **43** and heat-reflecting groove **44** may be circular elements arranged coaxially with communication axis z-z. Conductor **45** may be configured to transmit electricity to electric resistor **43** for conversion into heat energy **32B**. Groove **44** may include a concave shape extending proximally into housing **33** to contain resistor **43**, and the shape may include a distal surface configured to reflect heat energy **32B** toward skin **2**. In this configuration, heat signal **32B** may be output in signal direction SD as an amount of heat transferred to skin **2** by resistor **43**. Aspects of heat signal **32B** may be modified. For example, the size, shape, and/or exterior coating of resistor **43** or groove **44** may be configured to modify the intensity of heat energy **32B**.

(57) Shock generator element **48** may be configured to communicate an electrical energy **32C** to the brain through nerves associated with skin **2**. As shown in FIG. 4C, shock generator element **48** may be an electroshock generator that converts electricity from power source **66** into an electrical shock recognizable by electricity-sensitive receptors, such as the muscles and pain receptors located in the dermis layer of skin **2**. For example, energy generator element **48** may include at least two electric contacts **49**, a conductor **50**, and an insulating material **51**. The conductors **50** may be metallic rods or wires extending distally from controller **34**. Insulating material **51** may be an epoxy surrounding each conductor **50**. Each contact **49** may include a discharge shape located on the distal-most end of one of conductors **50**. In this configuration, controller **34** may direct electricity through conductors **50**, and into the discharge shape of contact **49**, allowing electricity to flow through skin **2** between the contacts **49** to output electrical energy **32C**.

(58) As shown in FIG. 3B, the electrical contacts **49** may be spaced apart in a radial pattern coaxial with communication axis z-z. Any number of contacts **49** may be used, in any geometrical and/or spatial configuration. Insulating material **51** may be used to define and maintain the spacing. As shown, insulating materials **51** and **46** may be the same material, such as an epoxy. Four contacts

49 are shown in FIG. **3B**, for example, as being arranged in two pairs. Aspects of electrical energy **32C** may be modified. For example, the arrangement of contacts **49** may be changed; and/or the size of or spacing between each contact **49** changed to modify the intensity of energy **32C**.

(59) Pressure generator element **52** may be configured to communicate a pressure energy **32D** to the brain through nerves associated with skin **2**. As shown in FIG. **4D**, pressure generator element **52** may be an electroacoustic transducer that converts electricity from power source **66** into a sound wave recognizable by pressure receptors of skin **2**, such as the Pacinian corpuscle. For example, pressure generator element **52** may include a cone **53**, a voice coil **54**, and a magnet **55**. In this configuration, controller **34** may direct electricity into voice coil **54** for interaction with magnet **55**, causing movements of cone **53** that generate the pressure energy **32D** in signal direction **SD**.

(60) As shown in FIGS. **3B** and **4D**, cone **53** may have a frustoconical shape that is coaxial with communication axis **z-z**. An outer edge of cone **53** may be attached an interior surface of housing **33**, and an inner edge of cone **53** may be attached to voice coil **54**, which may be coupled to controller **34** and power source **66** by one or more conductors. As shown, coil **54** may have a circular shape, and generator elements **36**, **42**, and **48** may be located in the interior of said shape. Aspects of pressure energy **32D** may be modified. For example, cone **53** and/or voice coil **54** may include a surround, a spider, a secondary frame, or any other structures configured to modify signal responsiveness; the strength of magnet **55** may be varied; and/or controller **34** may include an amplifier configured to modify an intensity of pressure energy **32D**.

(61) Different generator element types also may be used to communicate signals to the skin with different energies **32**, and/or different combinations of energies **32**. For example, the plurality of generators **31** may be modified to vary individual or combined outputs of energies **32A**, **32B**, **32C**, and **32D**; and/or include additional generator elements configured to output additional signals to skin **2**, including optical signals, magnetic signals, and/or any physically recognizable signals. Any type of generator element may be used and likewise coaxially arranged according to FIGS. **3A** through **4D**.

(62) Additional aspects of an exemplary processing unit **60** are now described with reference to FIG. **5**. As shown, processing unit **60** may be configured to receive input data **80** from a data source **81** and output control signal **82** and/or electricity to each controller **34** via conductors **27**, causing activation of one or more energy generators **31**. For example, processing unit **60** of FIG. **5** includes a housing **61**, a data transceiver **62**, one or more processors **63**, a memory **64**, a communication bus **65**, and a power source **66**.

(63) Data source **81** may include any combination of local and/or remote data sources. For example, source **81** may include a local sensor that is located in one of communication bays **25** and configured to send input data **80** to unit **60** using conductors **27** and/or bus **65**. This configuration may allow for closed loop communications in which energy signal **90** is based on data from the local sensors. For example, the local sensor may generate the input data **80** based on chemical and/or physical outputs related to skin **2**.

(64) Data source **81** also may include a remote data source in constant communication with processing unit **60** via data transceiver **62**, such as a remote sensor configured to send input data **80** to processing unit **60** with data transceiver **62** over a wired or wireless connection. This configuration may allow for open loop communications in which energy signal **90** is based on data from the local sensor and/or the remote sensor.

(65) Any number and type of local sensors may be used to generate input data **80**, and the sensor(s) may be located at any position on or relative to energy transceiver **10**. In the healthcare setting, for example, one local sensor may include a personal health tracker (e.g., a Fitbit® or an iWatch®) configured to generate input data **80** based on chemical and/or physical outputs of the wearer (e.g., heart rate, temperature), and communicate input data **80** to data transceiver **62** at regular intervals (e.g., once per second or once per minute).

(66) Housing **61** may contain the elements of processing unit **60**, and/or provide a means for

removing processing unit **60** from body **2**, allowing for easy repairs and upgrades. As shown in FIGS. **1B** and **5**, for example, exterior surfaces of housing **61** may be snap-fit with interior surfaces of compartment **23** so that the distal surface of processing unit **60** is maintained against the proximal surface of compartment **23**. For example, the exterior surfaces of housing **61** of may include protrusions biased outwardly along the X-X and Y-Y axes, and the interior surfaces of compartment **23** may include grooves configured to receive said protrusions.

(67) Transceiver **62** may include any wired or wireless communication technology configured to receive input data **80** from any data source(s) **81**, such as Bluetooth, Wi-Fi, **33**, and the like. As shown in FIG. **5**, input data **80** may be generated with or stored on data source **81** and received with transceiver **62**. In a healthcare setting, for example, data source **81** may include at least one patient monitoring device configured to send input data **80** to a remote server at regular intervals (e.g., once per minute). Data **80** may include various measures regarding the patient, such as body temperature, pulse rate, respiration rate, and/or blood pressure. For example, transceiver **62** may be configured to retrieve and/or receive data **80** from the remote server at regular intervals (e.g., once per second or once per minute).

(68) Each control signal **82** may be received with input data **80**. Data transceiver **62** may be configured to relay the signals **82** to the one or more processors **63** and/or memory **64**.

Alternatively, processing unit **60** may be configured to generate each control signal **82** based on input data **80**. For example, memory **64** may include a signal generating program, and the one or more processors **63** may be configured to generate each control signal **82** with the program. In keeping with previous examples, the signal generating program may be configured to: analyze the input data **80** sent from data sources **81** including a patient monitoring device during an interval; generate symbol **92A** from the temperature and pulse rate, symbol **92B** from the respiration rate, and symbol **92C** from the blood pressure; and output a control signal **82** for communicating the symbols **92A**, **92B**, and **92C** to skin **2**.

(69) As shown in FIG. **5**, communication bus **65** may be configured to connect the one or more processors **63** and memory **64** to each generator **31**, such as to each controller **34**. Bus **65** may include electrical and/or optical connectors **67** located on and/or extending distally through housing **61**. For example, communication bus **65** may comprise a flexible circuit board including a proximal surface supporting elements of processing unit **60**, and a distal surface including an electrical and/or optical network extending from power source **66** to the connectors **67**. Any type of network may be used, such as a mesh network. Connectors **67** may be engageable with corresponding connectors of conductors **27** to provide at least one pathway for outputting control signal **82** from processing unit **60** to one or more generators **31**, and/or electricity from power source **66** to one or more generators **31**. Control signal **82** may include electrical and/or optical signals. For example, control signal **82** may include a string of output commands for each generator **31**, and the entire string may be output to each generator **31** utilizing the electrical and/or optical signals, adding resiliency, in which the optical signals may be utilized for faster transmission.

(70) As described above, the snap-fit connection between housing **61** and compartment **23** may place connectors **67** in communication with conductors **27**, and maintain that communication over time, allowing for continuous output of control signals **82** from processing unit **60** and/or electricity from power source **66**. A cover element may be attached to the proximal surface **24** of body **20** to seal processing unit **60** within compartment **23**, and/or reinforce or supplant the snap-fit connection between housing **61** and compartment **23**. For example, the cover may include a graphic design, a textual element, a writing surface, and/or like decorative feature. As a further example, the cover may provide a mounting surface for other technologies, such as an antenna, signal amplifier, and/or supplemental data transceiver.

(71) Power source **66** may include any means for supplying electricity to processing unit **60** and/or the plurality of generators **31** (e.g., to each controller **34**). As shown in FIG. **5**, power source **66**

may include a rechargeable battery, such as a lithium ion battery, chargeable by connection to an external power source, such as a wall outlet. Power source **66** may include power generation technologies. For example, a proximal surface of power source **66** may include a power generator, such as photovoltaic cells configured to charge the battery. As shown in FIG. 5, power source **66** also may include an optical energy source, such as a laser generator that is powered by power source **66** and configured to output optical energy to one or more generators **31** via optical pathways defined by communication bus **65** and conductors **27**.

(72) Aspects of attachment element **70** are now described with reference to FIG. 2C. As shown, attachment element **70** may be configured to maintain a position of tissue interface **30** against or adjacent skin **2**. For example, element **70** may include an adhesive, elastic, and/or fastening element configured to apply a maintaining force in signal direction SD. In FIG. 2C, element **70** includes a proximal surface **72** adhered to the distal surface **24** of body **20**, and a distal surface **74** adherable with skin **2**. Distal surface **74** of element **70** may include a biocompatible adhesive configured to apply the maintaining force.

(73) Attachment element **70** may be removably and/or semi-permanently attached to skin **2** by the biocompatible adhesive. For example, a first adhesive material may be used to attach the proximal surface **72** to distal surface **24**, and a second adhesive material may be used to attach distal surface **74** to skin **2**. As a further example, the first adhesive may be stronger so that energy transceiver **10** may be removed from skin **2** without separating surfaces **72** and **24**. Either the first or second adhesive material may be biocompatible, and may include anti-bacterial and/or moisture resistant coatings and/or compositions configured for prolonged contact with skin **2**. For example, at least the second adhesive material may be configured for contact with skin **2** during the entirety of a 4-hour, 8-hour, 12-hour, 24-hour shift, or longer shift. One or both adhesives also may be configured for semi-permanent contact with skin **2**, such as during the entirety of a multi-month or multi-year treatment period. For example, at least the second adhesive material may include medicinal coatings and/or compositions that promote prolonged or semi-permanent contact with skin **2** by time-releasing treatments configured to prevent or minimize contact-based injuries.

(74) Body **20** and/or attachment element **70** may be configured to boost the efficacy of energy signal **90** by minimizing and/or maintaining the distance between tissue interface **30** and skin **2**, allowing signal **90** to be communicated with less energy. For example, any of the one or more energies **32** may be output through body **20** and/or attachment element **70**. As shown in FIGS. 2B and 2C, attachment element **70** may include a plurality of openings **76**. Each opening **76** may be sized approximate to one of communication bays **25**, allowing the energies **32** to be output towards skin **2** in signal direction SD through openings **76**. For example, each opening **76** may have an inner diameter approximate to an outer diameter of the communications bay **25** or housing **33** for each generator **31**. As shown in FIG. 2C, attachment element **70** may have a thickness that allows tissue contact **39**, electrical resistor **43**, and/or electrical contacts **49** to contact skin **2** through opening **76** or be adjacent to skin **2** within opening **76**.

(75) Aspects of body **20** and/or attachment element **70** may direct and focus the energies **32**, making it easier for the brain to distinguish one output of energies **32** from another. In keeping with previous examples, body **20** and attachment element **70** of FIGS. 2B and 2C may be composed of base materials including an impact absorbing material configured to absorb any excessive vibrations of skin **2** caused by impact energy **32A**. One or both base materials may include an insulating material configured to direct heat energy **32B**, electrical energy **32C**, and pressure energy **32D** through openings **76** along axis Z-Z; and prevent transmission of energies **32B**, **32C**, and **32D** along axis X-X and Y-Y. For example, body **20** and element **70** of FIG. 2C may be configured to absorb any portion of energies **32** output incidentally in directions transverse to signal direction SD to promote signal distinction by limiting unwanted communications. As a further example, each opening **76** of attachment element **70** in FIG. 2C may have a reflective coating and/or a frustoconical interior shape centered about axis z-z to further focus the energies **32** towards skin **2**.

(76) As described herein, energy transceiver **10** may be operable to communicate energy signal **90** to skin **2** by outputting any energy **32**, such as impact energy **32A**, heat energy **32B**, electrical energy **32C**, and/or pressure energy **32D**, individually or together. For example, any energies **32A-D** may be used interchangeably or in combination to communicate any of the dots shown in FIG. **1A** as symbols **92A**, **92B**, and **92C**. As now described, aspects of each energy **32** may be modified to increase the complexity of signal **90**, and thus the amount of data transmitted therewith. Modifiable aspects may include energy type, energy intensity, output duration, scroll rate, symbol shape, and the like.

(77) Energy signal **90** may be communicated to skin **2** with energies **32**, individually or together. In FIG. **1A**, for example, each dot within first symbol **92A** may be output with impact energy **32A**; each dot within second symbol **92B** may be output with heat energy **32B**; and each dot within third symbol **92C** may be output with electrical energy **32C**. The energies **32** may be combined for additional emphasis. For example, the first symbol **92A** may be output with impact energy **32A** in response to a baseline measure, and output with a combination of impact energy **32A** and heat energy **32B** if the measure changes. The energies **32** also may be combined to enhance the penetration depth of energy signal **90**. For example, first symbol **92A** may be formed by first outputting pressure energy **32D** to activate a portion of the nerves associated with skin **2**, and second outputting heat energy **32B** to the activated nerves. Any individual dot may be similarly modified relative to any other dot.

(78) The intensity of energies **32** may be modified for emphasis. For example, processing unit **60** may be configured to output first symbol **92A** with impact energy **32A** at a first intensity level in response to a baseline measure, and a second intensity level to highlight signal **92A** if the measure changes. Output duration may be similarly modified. For example, the output duration of energies **32** may be instantaneous for normal measures, like a quick tap (e.g., about 100 ms); extended for abnormal measures, like a short hold (e.g., 500 ms to 1 s); or a combination thereof, as with Morse code. Scroll rate may be similarly modified. For example, symbols **92** may not be scrolled at all (i.e., a scroll rate of zero), and output duration may be used to communicate change over time by flashing symbols **92** off and on in a fixed position. As a further example, in the healthcare setting, the scroll rate may be based on an update schedule (e.g., one revolution per minute), and/or the output duration may be based on patient status (e.g., faster for more critical patients).

(79) Symbol shape also may be modified. The plurality of symbols **92** are shown as pip pattern shapes in FIG. **1A**, but any symbol shape may be used, particularly those amenable to dot-matrix representation. For example, the plurality of symbols **92** may include known Morse code, binary symbols, lines, and/or directional arrows that are scrolled across communication area **4** in communication direction **CD**. Alphanumeric symbols also may be communicated. For example, input data **80** may include a control signal **82** generated from a Twitter® feed, and the symbols **92** may include alphanumeric symbols for communicating the author, date, and content of each Tweet® contained in the feed. As a further example, input data **80** may include the subject and sender of an email, and the signal generating program included in memory **64** may be configured to: prioritize the email based on the sender; and generate a control signal **82** for outputting a set symbols **92** based on the subject, sender, and priority of the email. For example, first symbols **92** may be output with impact energy **32A** to communicate the subject and/or sender of prioritized emails in a shorthand notation, and at least one of heat energy **32B**, electrical energy **32C**, pressure energy **32D** to communicate the priority level of the shorthand notation.

(80) The resolution of tissue interface **30** may match or exceed the distinguishing capabilities of the nerves associated with skin **2**. For example, in the grid formation shown in FIG. **2B**, the resolution of tissue interface **30** may be measured as energy output per square inch, which may exceed the natural energy receptivity limits of the nerves associated with skin **2**. As shown, the resolution of interface **30** may be relative to the spacing between each bay **25**, the configuration of body **20** and/or attachment element **70**, and/or the intensity of energies **32**. The energy receptivity limits of

skin 2 may vary by location. For example, energy transceiver 10 may be attached to a portion of skin 2 located in a highly innervated or sensitive area, such as the face, allowing even more complex symbol shapes to be communicated.

(81) With sufficient resolution, tissue interface 30 may likewise be configured to output signal 90 to replicate image patterns and/or other sensory perceptions with energies 32, including any of the symbols described herein and even more complex interactions. As described herein, the multi-energy capabilities of energy transceiver 10 may be configured to layer energies 32 so as to communicate far more complex image patterns and/or sensory perceptions that would otherwise be possible by communicating with a single energy because of the natural receptivity limits of the nerves, and their tendency to become less receptive during prolonged exposures.

(82) Additional aspects of this disclosure are now described with reference to numerous additional examples of energy transceiver 10, including: an exemplary energy transceiver 110 shown conceptually in FIG. 6A; an exemplary energy transceiver 210 shown conceptually in FIG. 6B; an exemplary energy transceiver 310 shown conceptually in FIG. 6C; an exemplary energy transceiver 410 shown conceptually in FIG. 6D; an exemplary energy transceiver 510 shown conceptually in FIG. 7A; an exemplary energy transceiver 610 shown conceptually in FIG. 7B; an exemplary energy transceiver 710 shown conceptually in FIG. 7C; an exemplary energy transceiver 810 shown conceptually in FIG. 7D; and an exemplary energy transceiver 910 shown conceptually in FIGS. 9A-B.

(83) Each variation of transceiver 10, such as transceivers 110, 210, 310, 410, 510, 610, 710, 810, 910, may include elements similar to those of transceiver 10, but within the respective 100, 200, 300, 400, 500, 600, 700, 800, or 900 series of numbers, whether or not those elements are depicted in FIGS. 6A through 9B. Any aspects described with references to transceivers 110, 210, 310, 410, 510, 610, 710, 810, and 910 may be included within any variation of transceiver 10 described herein, each possible combination or iteration being part of this disclosure. For example, any variation of transceiver 10 may comprise any combination of the wearable aspects of transceivers 110, 210, 310, and 410; the contact-based aspects of transceivers 510 and 610; and the implantable aspects of transceivers 710 and 810; and/or any multi-signal aspects of transceiver 910.

(84) Additional wearable aspects are now described with reference to FIGS. 6A-D. As shown in FIG. 6A, energy transceiver 110 may include: a body 120 and a tissue interface 130, both shown conceptually with a dotted line; and an attachment element 170, shown conceptually as a sweat band. Any type of band may be used, such as a head band, an arm band, or a bandana. Body 120 may wrap around a portion of skin 2, such as circular portion of skin 2, like around a human forearm or forehead. As shown in FIG. 6A, body 120 may be mounted on attachment element 170; and tissue interface 130 may be mounted on a distal surface of body 120. For example, body 120 may be mounted on a distal surface of element 170; and tissue interface 130 may wrap around the circular portion of skin 2 with body 120, providing a curved rectangular communication area 4 and a semi-circular (e.g., less than 360°) or circular (e.g., 360°) communication direction CD for energy signal 90.

(85) Attachment element 170 (e.g., a sweat band) may be configured to maintain tissue interface 130 against or adjacent a portion of skin 2, such as against the arm or forehead, allowing energy signal 90 to be output in signal direction SD and/or scrolled around the head to communication area 4 in communication direction CD. For example, the band may include an elastic portion that pushes body 120 and tissue interface 130 distally toward skin 2 when placed around the circular portion of skin 2, i.e., when the sweat band of FIG. 6A is worn. As shown, the elastic portion may be proximal of energy transceiver 110, attached to a proximal surface of body 120, and configured to apply a circumferential maintaining force that maintains the position of interface 130 when element 170 is worn.

(86) As shown in FIG. 6B, energy transceiver 210 may include: a body 220 and a tissue interface 230, both shown conceptually with a dotted line; and an attachment element 270, shown

conceptually as a baseball cap. Any cap, hat, helmet, or like headwear may be used. Body **220** may wrap around a circular portion of skin **2** including the forehead and/or scalp. As shown in FIG. **6B**, body **220** may be mounted on attachment element **270**; and tissue interface **230** may be mounted on a distal surface of body **220**. For example, body **220** may be mounted on a distal surface of element **270**; and tissue interface **230** may wrap around the circular portion of skin **2** with body **220**, providing a semi-circular (e.g., less than 360°) or circular (e.g., 360°) communication area **4** and communication direction CD for energy signal **90**. As a further example, body **220** and tissue interface **230** may have a semi-spherical shape covering interior surfaces of cap **270** for output of energy signal **90** to a semi-spherical communication area **4** of skin **2** in any communication direction(s) CD.

(87) Similar to attachment element **170** of FIG. **6A**, attachment element **270** (e.g., a cap) also may include an elastic or non-elastic portion configured to maintain tissue interface **230** against or adjacent a portion of skin **2**, such as against the forehead, allowing energy signal **90** to be output in signal direction SD and/or scrolled around the head in communication direction CD. For example, the elastic or non-elastic portion may push body **220** and tissue interface **230** distally toward skin **2** when placed around the circular portion of skin **2**, i.e., when the cap of FIG. **6B** is worn. As shown, the elastic or non-elastic portion may be proximal of energy transceiver **210**, attached to a proximal surface of body **220**, and configured to apply a circumferential maintaining force that maintains the position of interface **230** when attachment element **270** is worn. For example, the elastic or non-elastic portion may comprise a tension fastening mechanism, such as any snaps, Velcro®, or other typically found on headwear.

(88) As shown in FIG. **6C**, energy transceiver **310** may include a body **320** and a tissue interface **330**, both shown conceptually a dotted line; and an attachment element **370**, shown conceptually as a sock. Any tube-like garment may be used, including gloves, shoes, stockings, and the like. Body **320** may wrap around a circular portion of skin **2**, such as around a leg. As shown in FIG. **6C**, body **320** may be mounted on attachment element **370**; and tissue interface **330** may be mounted on a distal surface of body **320**. For example, body **320** may be mounted on a distal surface of element **370**, and tissue interface **330** may wrap around the circular portion of skin **2** with body **320**, providing a semi-circular (e.g., less than 360°) or circular (e.g., 360°) communication area **4** and direction(s) CD.

(89) Similar to above, attachment element **370** (e.g., a sock) may include an elastic layer configured to maintain tissue interface **330** against or adjacent a portion of skin **2**, such as against the leg, allowing energy signal **90** to be output in signal direction SD and/or scrolled around the head in communication direction CD. For example, the elastic layer may push body **320** and tissue interface **330** distally toward skin **2** when placed around the circular portion of skin **2**, i.e., when the sock of FIG. **6C** is worn. As before, the elastic layer may be proximal of energy transceiver **310**, attached to a proximal surface of body **320**, and configured to apply a circumferential maintaining force that maintains the position of interface **330** when attachment element **370** is worn.

(90) As shown in FIG. **6C**, transceiver **310** may be removably attached to attachment element **370**, and thus operable with a plurality of elements **370**, such as plurality of socks or other tube-like garments that typically become soiled during use. For example, element **370** may include a pouch configured to receive and secure body **320**, orient tissue interface **330** toward skin **2**, and/or maintain the position tissue interface **330** on or adjacent skin **2**. As a further example, the elastic layer and/or the pouch may include an opening, and portions of body **320** may be engageable with (e.g., snap fit into) the opening to further maintain interface **330**.

(91) As shown in FIG. **6D**, for example, energy transceiver **410** may include a body **420** and a tissue interface **430**, both shown conceptually with a dotted line; and an attachment element **470**, shown conceptually as a compression garment. Any type of compressive garment may be used, such as those made by Under Armour®. Body **420** of FIG. **6D** may wrap around a portion of skin

2, and be composed of an impacting absorbing material (e.g., foam) configured to dissipate external impact forces directed toward the skin 2. For example, body 420 may be a thigh pad used in hockey or American football, a shin guard used in soccer, or any other type of protective pad with a distal surface that is desirably maintained against skin 2. Similar to above, body 420 may be mounted on attachment element 470; and tissue interface 430 may be mounted on a distal surface of body 420. For example, body 420 may be mounted on a distal surface of element 470; and tissue interface 430 may wrap around the circular portion of skin 2 with body 420, providing a curved communication area 4 and direction CD for signal 90.

(92) Attachment element 470 (e.g., the compression garment) may include an elastic weave configured to maintain tissue interface 430 against or adjacent a portion of skin 2, such as against the arm or forehead, allowing energy signal 90 to be output in signal direction SD and/or scrolled around the head to communication area 4 in communication direction CD. For example, the elastic weave may push body 420 and tissue interface 430 distally toward skin 2 when placed around the circular portion of skin 2, i.e., when the compression garment of FIG. 6D is worn. In this example, the elastic weave may be attached to body 420, and configured to apply a circumferential maintaining force that maintains the position of interface 430 when attachment element 470 is worn.

(93) Similar to above, energy transceiver 410 may be removably attached to attachment element 470, and thus operable with a plurality of elements 470, such as plurality of compressive garments. For example, impact absorbing body 420 may be mounted in a pocket of attachment element 470, and tissue interface 430 may be mounted on a distal surface of impact absorbing body 420, such as in a distal compartment of body 420. Transceiver may be a game-time accessory. For example, as shown in FIG. 6D, signals 90 may comprise a plurality of arrows scrolled along communication direction CD to communicate movements to the user. In this example, the user may be trained to move in a particular direction (e.g., left or right) and intensity (e.g., slow or fast) based on the output of energy signal 90 and the particular combination of energies 32 associated therewith.

(94) Although not shown in FIGS. 6A-D, attachment elements 170, 270, 370, and 470 may include any adhesive and/or energy focusing elements, including those described above. For example, any aspects of attachment element 70 of FIG. 2C may be combined with any aspects of attachment elements 170, 270, 370, or 470 of FIG. 6A-D to further maintain a position of tissue interface 430 relative to skin 2 and/or focus the energies 32 according to aspects of this disclosure. Aspects of any attachment elements may be combined and operable together. For example, attachment element 470 may include an opening, body 420 may be snapped into the opening, and a second attachment element (e.g., a biocompatible low-tack adhesive) may be configured to further fix the position of interface 430 relative to skin 2 during rigorous physical activity, such as running.

(95) As described above, aspects of energy transceivers 110, 210, 310, and 410 may be included with any wearable item, giving aspects of this disclosure incredible breadth. For example, aspects of any of attachment elements 170, 270, 370, and 470 may be integrated into any wearable item including any type of band, strap, or like item including any combination elastic and/or non-elastic layers or portions. Exemplary attachment elements may include: bandages, wherein the tissue interface may be located on a distal surface of a skin-attachment portion; belts, wherein the tissue interface may be located on a distal surface of the belt; bras, wherein the tissue interface may be located on a distal surface of a bra strap; earrings, wherein the tissue interface may be located on a distal surface of an earring front or back; pants, wherein the tissue interface may be located on a distal surface of a waste line or seam; rings, wherein the tissue interface may be located on an interior distal surface; shirts, wherein the tissue interface may be located on a distal surface of a neckline; underwear, wherein the tissue interface may be located on a distal surface of the legs or waistline; watches, wherein the tissue interface may be located on a distal surface of the watch strap; and any known or obvious variation of the same.

(96) Aspects of transceivers 10, 110, 210, 310, and 410 may be likewise included on any non-

wearable object with a distal surface that is desirably maintained against skin 2 during use by application of an external force, such as a gravity force, a gripping force, or other externally applied maintaining force. Additional external force-based aspects are now described with reference energy transceiver **510** of FIG. 7A and energy transceiver **610** of FIG. 7B.

(97) As shown in FIG. 7A, energy transceiver **510** may include: a body **520** and a tissue interface **530**, both shown conceptually with a dotted line; and an attachment element **570**, shown conceptually as a shoe or a shoe insert. Any type of footwear and/or foot support with equivalent surfaces may be used. Body **520** may include a surface contoured for placement against skin 2, such as an underside of a foot. As shown in FIG. 7A, body **520** may be mounted on attachment element **570**; and tissue interface **530** may be mounted on or embedded in a distal portion of body **520**, allowing gravity to at least partially maintain interface **530** against or adjacent skin 2, and providing a foot-shaped communication area **4** and communication direction CD for energy signal **90**.

(98) Attachment element **470** may additionally comprise any tensioning elements configured apply a maintaining force that maintains the position of interface **530** when attachment element **570** is worn, such as shoelaces, Velcro, pumping mechanisms, elastic straps or structures, and the like. As a further example, attachment element **570** may be composed of an impact absorbing material, such as a polymeric material configured to distribute forces around body **520** when walking or running; and include bolster shapes contoured to further maintain tissue interface **530** by limiting lateral movements of the foot relative thereto.

(99) Accordingly, energy signal **90** may be communicated to the communication area **4** of skin 2 by tissue interface **530** in any communication direction CD with any combination of energies **32**. As shown in FIG. 7A, energy signal **90** may include a plurality of directional shapes (e.g., the arrows of FIG. 7A) flashed and/or scrolled in a linear direction to communication directional movements. The directional shapes may be responsive to directional data. For example, transceiver **510** may be configured to receive the directional data from one or more sources (e.g., GPS signals), determine communication direction CD based on the directional data, and scroll energy signal **90** across skin 2 as directional shapes scrolling along communication direction CD to compel movement of the user in a direction.

(100) In keeping with above, transceiver **510** also may be configured to determine an importance measure based on the directional data, and communicate energy signal **90** with a particular combination of energies **32** and/or at a particular scroll rate based on the importance measure to direct a movement aspect, such as pace or direction. In the healthcare setting, for example, the directional data may include a vital sign of a patient and the GPS location of the patient; and transceiver **510** may determine the scroll rate based on the vital sign, allowing energy signal **90** to guide a healthcare provider toward the patient at walking pace appropriate for the condition of the patient. For example, energy signal **90** may be communicated a faster scroll rate with high intensity energies **32** to alert the provider to run if needed.

(101) As shown in FIG. 7B, for example, energy transceiver **610** may include a body **620** and a tissue interface **630**, both shown conceptually with a dotted line; and an attachment element **670**, shown conceptually as a grip panel attached to the grip of a gun. Any type of gun may be used, including the handgun with a pistol grip and any other type of gun with similar surfaces that are gripped during use. In FIG. 7A, tissue interface **630** may be located on a distal surface of element **670** that is typically pushed toward a portion of skin 2 of a hand by a grip force applied by the hand during use, providing a regular or irregular shaped communication area **4**. Attachment element **630** may further maintain interface **630** by limiting movements the hand. For example, element **630** may be 3D printed based on a scan of the hand to include an outwardly curving surface shaped that maintains interface **630** against skin 2 by limiting movements of the hand relative to tissue interface **430** when gripped.

(102) Accordingly, energy signal **90** may be communicated to the communication area **4** of skin 2

by tissue interface **630** in any communication direction CD with any combination of energies **32**. Aspects of energy signal **90** may be responsive to data, as with previous examples. For example, as shown in FIG. 7B, attachment element **670** (e.g., the gun) may include a sight, and energy signal **90** may include at least one decisional shape flashed and/or scrolled to communicate a status associated with gun based on a position of the sight. For example, energy transceiver **610** may include an elevation or motion sensor, and signal **90** may be output to skin **2** as a first shape (e.g., a circle) with a first energy (e.g., any combination of energies **32A-D**) whenever the sensor indicates that the sight of the gun has been raised with the safety off, alerting the user to a status of the gun. As a further example, energy transceiver **10** and/or the gun may be configured to determine whether the sight is aligned with a specific target, and output signal **90** as a second shape (e.g., an X shape) with a second energy (e.g., any combination of energies **32A-D**), alerting the user to a status of the target.

(103) Any individual or combined aspects of energy transceivers **10**, **110**, **210**, **310**, **410**, **510**, and **610** may likewise be included on any non-wearable object with a distal surface that is desirably maintained against skin **2** during use by application of an external force, such as a gravity, a gripping force, or other externally applied maintaining force. For example, aspects of tissue interfaces **510** and **610** may likewise be included on a distal surface of any load bearing surface of any type of attachment element. For example, aspects of attachment element **570** or **670** of FIGS. 7A-B alternatively may include a bar, a chair, a handle, a floor, a rope, a wall, or any like object with a skin facing surface that is generally maintained against skin **2** during use; and aspects of issue interfaces **530** or **630** of FIGS. 7A-B alternatively may be mounted therewith so that energy signal **90** may be output to skin **2** whenever the alternative attachment element **570** or **670** is used.

(104) Additional implantable aspects are now described with reference to energy transceiver **710** of FIG. 7C and energy transceiver **810** of FIG. 7D. As shown in FIG. 7C, for example, energy transceiver **710** may include a body **720** and a tissue interface **730**, both shown conceptually with a dotted line; and an attachment element **770**, shown conceptually as a portion of a bone plate. Any type of bone plate or other implantable object may be used. Attachment element **770** of FIG. 7C includes a proximal bone-facing surface and a distal skin-facing surface. The bone-facing surface may be maintained against the bone by any combination of adhesives, screws, wires, and/or other bone fixation technologies. The skin-facing surface may maintain the position of tissue interface **730** relative to the bone, allowing for movement of skin **2** relative to interface **730**. For example, tissue interface **730** may be mounted in a compartment on the skin-facing surface.

(105) As above, energy signal **90** may be communicated to the communication area **4** of skin **2** by tissue interface **730** in any communication direction CD with any combination of energies **32**. As shown in FIG. 7C, signal **90** may include any combination of shapes moving in any communication direction CD, including any combination of shapes and/or directions, any of which may be flashed and/or scrolled with any energies **32**. In contrast to above, energy signal **90** of FIG. 7C may be output toward the underside of skin **2**, allowing for more direct communication with nerves associated with skin **2**. Aspects of transceiver **710** and energy signal **90** may be modified according to the implanted location of attachment element **770**. For example, tissue interface **730** may be embedded in an attachment element **770** sized for placement against a radius or finger bone wherein the distance between skin **2** and bone is minimal, allowing signal **90** to be communicated with less energy. As a further example, interface **730** may be mounted on an attachment element **770** sized for placement against a radius or ulna, wherein the distance between skin **2** and the bone is larger.

(106) Similar to above, aspects of body **720** and/or attachment element **770** may direct and focus the energies **32**, making it easier to distinguish one output of energies **32** from another and/or prevent the energies **32** from being output to bone. Alternatively, all or portion of the energies **32** may be output toward the bone-facing surface of element **770** to communicate signals and/or apply treatments to the bone. For example, the energies **32** may be output through body **720** and/or

attachment element **770** in a proximal and/or distal direction, such as through a plurality of openings extending through element **770**. As a further example, the distal surface of body **720** may include a first tissue interface **730** and/or the bone-facing surface of body **730** may include a second tissue interface **730**, allowing a corresponding set of first and/or second energy signals **90** to be toward in a first direction toward skin **2** and/or a second direction toward the bone.

(107) As shown in FIG. 7D, for example, energy transceiver **810** may include a body **820** and a tissue interface **830**, both shown conceptually with a dotted line; and an attachment element **870**, shown as a biocompatible outer surface layer surrounding body **820**. Any type of biocompatible material or containing structure may be used. In this example, attachment element **870** may comprise a tissue in-growth promoting exterior layer that maintains the orientation and/or position of tissue interface **30** over time by interacting with living tissue. For example, element **870** may be composed of a polymeric material, such as a variant of polyether ether ketone (or “PEEK”); and/or include an outer surface textured to promote tissue ingrowth.

(108) Energy signal **90** may be output from tissue interface **830** as above. As shown in FIG. 7D, tissue interface **830** may be oriented so that the signals **90** are output through attachment element **870** and toward a communication area **4** under skin **2** in a signal direction SD. For example, aspects of body **820** and/or attachment element **870** may direct and focus the energies **32** toward discrete areas **4** on the underside of skin **2**, making it easier for the brain to distinguish one output of energies **32** from another and/or preventing the energies **32** from being output other living portions, such as bone or muscle. Alternatively, all or portion of the energies **32** may be output simultaneously from proximal and distal sides of element **870** to communicate with nerves associated with skin **2** and the other living portions. For example, the energies **32** may be output through body **820** and/or attachment element **870** in either direction through a plurality of openings extending therethrough. Also similar to above, transceiver **810** may include a first interface **830** disposed opposite a second interface **830**, allowing for output of a corresponding set of first and second energy signals **90**.

(109) Additional aspects are now described with reference to energy transceiver **910** of FIGS. **8A** and **8D**, demonstrating that any variation of transceiver **10** described herein may be configured to output a plurality of signals **90**. For example, energy transceiver **910** may be configured to output a plurality of energy signals **90** in a signal direction SD toward skin **2**. As shown in in FIGS. **8A** and **8B**, transceiver **910** may output a first signal **990.sub.1** in a first divided area or band **924.sub.1**, and a second signal **990.sub.2** in a second divided area or band **924.sub.2**. Each signal **990.sub.1** and **990.sub.2** may include a plurality of symbols. In FIG. **8A**, for example, the symbols include dots made visible through an exemplary cut-out in transceiver **910** as they would be communicated to skin **2**, similar to FIG. **1A**. Likewise, some of the generators **931** are shaded in FIG. **8A** indicate output of energies **32**, similar to FIG. **2B**.

(110) Transceiver **910** may comprise: a body **920**; a tissue interface **930**; a processing unit **960**; and an attachment element **970**. Similar to above, body **920** may contain elements of transceiver **10** within a flexible biocompatible base material that is conformable against skin **2**, and maintainable against skin **2** for prolonged and/or semi-permanent durations. As shown in FIGS. **8A** and **8B**, body **920** may have a length extending along a longitudinal axis X-X, a width extending along a lateral axis Y-Y, and a thickness extending along a proximal-distal axis Z-Z, similar to body **20** of FIGS. **2A-C**. The length, width, and/or thickness of body **920** may be compatible with a curved portion of skin **2**, as in FIG. **8A**, where body **920** is curved along axis X-X. For example, body **920** may be curved and/or wrapped around any body shape, such as a human forearm, a human shin, and/or portions of a human torso.

(111) As also shown in FIGS. **8A** and **8B**, body **920** may define a proximal surface **922**, a distal surface **924**, a distal compartment **926**, and an interior conduit **928**. The proximal surface **922** may include a cover **923** mounted thereto. Cover **923** may include a graphic design, a textual element, a writing surface, and/or like decorative feature. As shown in FIG. **8B**, a distal surface of cover **923**

may include a first attachment element (e.g., a first Velcro strip) engageable with a second attachment element (e.g., a second Velcro strip) on the proximal surface **922**, allowing cover **923** to be switched-out as needed. The second attachment element also may attach body **920** to another object such as the inside of a garment.

(112) Tissue interface **930** may be similar to any variation of tissue interface **30** described herein. As shown in FIG. **8B**, tissue interface surface **930** may be mounted in the distal compartment **926** of body **920**, and include plurality of energy generators **931** directed toward skin **2**. Each generator **931** may be similar to generators **31** described above. For example, each generator **931** may be operable with processing unit **960** to output energies **32** individually and/or in combination in a signal direction SD; and contained with base material **933** (e.g., epoxy) that directs and/or focusses energies **32** in the signal direction SD. Each generator **931** may likewise include a plurality of generator elements arranged (e.g., coaxially) to output their respective energies **32** in approximately the same direction along an axis z-z, making the outputs interchangeable. As before, the energies **32** may include impact energy **32A** (e.g., FIG. **4A**), heat energy **32B** (e.g., FIG. **4B**), shock energy **32C** (e.g., FIG. **4C**), pressure energy **32D** (e.g., FIG. **4D**); and/or any like energies.

(113) In contrast to above, the plurality of generators **931** may be arranged into a plurality of divided areas or bands. As shown in FIG. **8B**, the width of body **920** along lateral axis Y-Y may include the first band **924.sub.1** of generators **931**, which may extend around the length of body **920** along a first longitudinal axis X.sub.1-X.sub.1 of transceiver **910**; and the second band **924.sub.2** of generators **931**, which may extend around the length of body **20** along a second longitudinal axis X.sub.2-X.sub.2. The generators **931** located in first band **924.sub.1** may be configured to output first signal **990.sub.1**, and the generators **931** located in second band **924.sub.2** may be configured to output second signal **990.sub.2**. To enhance distinguishability, an interior portion of compartment **926** and/or base material **933** may physically separate first band **924.sub.1** from second band **924.sub.2**, as in FIG. **8B**.

(114) As shown in FIG. **8A**, processing unit **960** may be configured to: receive first input data **980A** from a first data source **981A**; receive second input data **980B** from a second data source **981B**; and output a control signal **982** and/or to electricity to generators **931**, causing various combinations of said generators **931** to output first signal **990.sub.1** and second signal **990.sub.2**. For example, processing unit **960** of FIGS. **8A** and **8B** may include any elements of processor **60** of FIG. **5**, such as transceiver **62**, one or more processors **63**, memory **64**, communication bus **65**, and power source **66**. Each of these elements may perform a similar function within processing unit **960**. Similar to above, one or more wired and/or wireless connections (e.g., such as conductors **27**) may extend between processing unit **960** and each generator **931**.

(115) Attachment element **970** may maintain a position of tissue interface **930** against or adjacent skin **2**. As shown in FIG. **8B**, attachment element **970** may be proximal of tissue interface **930**, and configured to maintain the position of interface **930** by applying a distally-directed force to body **920**. The distally-directed force may press tissue interface **930** against skin **2**, and/or cause portions of interface **930** to conform against a curvature of skin **2**. As also shown in FIG. **8B**, attachment element **970** may include a strap **972** extending through an interior conduit **928** of body **920**. Strap **972** may apply the distally-directed force. For example, strap **972** may be composed of a resilient material (e.g., metal) having a cross-sectional shape (e.g., a semi-circular shape) that maintains body **920** in either an elongated configuration (e.g., FIGS. **2A-C**) or a curved configuration (e.g., FIG. **8A**), like a slap bracelet.

(116) Attachment element **970** also may apply the distally-directed force by applying a tensile force to strap **972**. As shown in FIG. **8A**, a first end **973** of strap **972** may extend from one end of conduit **928**, a second end **977** of strap **972** may extend from another end of conduit **928**, and the tensile force may be imparted by removably attaching ends **973** and **974**. For example, a proximal surface of the first end **973** may include a first attachment element (e.g., a first Velcro strip), a distal surface of second end **974** may include a second attachment element (e.g., a second Velcro strip), and the

first and second attachment elements may be overlapped to impart the tensile force. Any type of attachment element may be used to attached ends **973** and **974**, including buckles, ratchets, and the like. In some aspects, band **972** may be an elastic band, and ends **973** and **974** may be permanently attached together.

(117) Processing unit **960** may be removably attached to transceiver **910**, allowing for easy repairs and upgrades. As shown in FIG. **8A**, processing unit **960** may be attached to a distal surface of the first end **973** of strap **972**, and connected to tissue interface **930** by one or more conductors. For example, similar to conductors **27** described above, the conductors may include a network that is located in distal compartment **926** with tissue interface **930**, and configured to transmit power and/or control signals between processing unit **960** and generators **931**. As shown in FIG. **8A**, a distal surface of processing unit **960** may include one or more sensors **968**, and attachment element **970** may be configured to maintain a position of the one or more sensors **968** one or adjacent to skin **2**, allowing characteristics of the user to be monitored and/or output with processing unit **960**.

(118) Signals **990.sub.1** and **990.sub.2** may be similar to signal **90** of FIG. **1A**. For example, first signal **990.sub.1** may include a plurality of first symbols output in first band **924.sub.1**, and second signal **990.sub.2** may include a plurality of second symbols output in second band **924.sub.2**. As shown in FIGS. **8A** and **8B**, each of first and second symbols and/or dot may be associated with different data. For example, in the healthcare setting, first signal **990.sub.1** may include first symbols associated with a first patient, and each first symbol may be associated with a vital sign for the first patient; whereas second signal **990.sub.2** may include second symbols associated with a second patient, and each second symbol may be associated with a vital sign for the second patient, allowing the user to simultaneously monitor the first and second patients with transceiver **910**.

(119) As shown in FIG. **8A**, first signal **990.sub.1** may be scrolled around first band **924.sub.1** by outputting energies **32** toward skin **2** in signal direction **SD**, and moving the output across skin **2** in a first communication direction **CD.sub.1**; and second signal **990.sub.2** may be scrolled around second band **924.sub.2** by outputting energies **32** toward skin **2** in signal direction **SD**, and moving the output across skin **2** in a second communication direction **CD.sub.2**. Each signal **990.sub.1** and **990.sub.2**, and/or each first or second symbol included therein, may be configured for increased complexity, allowing more data to be transmitted therewith. In keeping with the previous healthcare example, each signal **990.sub.1** and **990.sub.2** may be scrolled in one of communication directions **CD.sub.1** or **CD.sub.2** at a scroll rate associated with a vital sign of the respective first and second patients (e.g., pulse rate); each first and second symbol may be associated with another vital sign for said first and second patients (e.g., body temperature, pulse rate, respiration rate, and/or blood pressure); and the first and second symbols may be output with different combinations of energies **32** to communicate different aspects the vital signals (e.g., an increase or decrease in body temperature, pulse rate, respiration rate, and/or blood pressure).

(120) Although shown as having two divided areas (e.g., first band **924.sub.1** and second band **924.sub.2**) configured to output two different energy signals (e.g., first signal **990.sub.1** and second signal **990.sub.2**), transceiver **910** may include any number of divided areas having any shape. For example, the width of body **920** may accommodate a plurality of divided areas, at least one tissue interface **930** may be located in each divide area, and attachment element **970** may be configured to maintain each tissue interface **930** against a different portion of skin **2**. For example, body **920** of FIG. **8A** may accommodate a plurality of bands spaced apart along a length of a limb (e.g., a forearm), and each band (e.g., similar to bands **924.sub.1** and **924.sub.2**) may output a different energy signal (e.g., similar to signals **990.sub.1** and **990.sub.2**) based on input data from a different data source (e.g., similar to sources **981.sub.1** and **981.sub.2**). In a healthcare setting, each data source may include a patient monitoring device, allowing the user to simultaneously monitor a plurality of different patients with transceiver **910**.

(121) Various methods associated with transceiver **10** are now described, including methods of

operating transceiver **10**. Aspects of each method may be used with any variation of transceiver **10** described herein, such as transceivers **110**, **210**, **310**, **410**, **510**, **610**, **710**, **810**, and **910** described above. For ease of description, aspects these methods are now described with various references to these exemplary transceivers, including numerous references to energy transceiver **10**. Unless claimed, these references are exemplary and non-limiting.

(122) As shown in FIG. **9**, an exemplary method **1000** may comprise: receiving, with processing unit **60**, input data **80** for a communication device **10** including a tissue interface **30** maintainable on or adjacent skin **2**, the interface **30** including a plurality of energy generators **31**, each generator **31** being operable to output a plurality of energies **32** in a signal direction SD toward skin **2** (a receiving step **1020**); and operating, with processing unit **60**, the plurality of energy generators **31** to communicate with nerves associated with the skin **2** by outputting one or more energies (e.g., any of energies **32A**, **32B**, **32C**, and **32D**) of the plurality of energies **32** in the signal direction SD based on input data **80** (an operating step **1040**).

(123) Receiving step **1020** may comprise receiving input data **80** from one or more data sources **81**. For example, the one or more data sources **81** may include at least one of patient monitoring device, a remote server, and a sensor. In this example, receiving step **1020** may comprise receiving input data **80** from the one or more data sources **81** at regular intervals, and operating step **1040** may comprise outputting the one or more energies based on the input data **80** received during each regular interval.

(124) Input data **80** may include a control signal **80**, and operating step **1040** may comprise outputting the one or more energies based on the control signal **82**. Alternatively, method **1000** may comprise: generating, with processing unit **60**, control signal **82** based on input data **80**, wherein operating step **1040** may comprise outputting the one or more energies based on control signal **82**. For example, the generating step **1030** may include associating the input data **80** with a plurality of symbols **92**, and operating step **1040** may comprise communicating the symbols **92** with the one or more energies. In this example, the input data **80** may include measurements (e.g., vital signs of a patient), and each symbol may be associated with one or more of the measurements (e.g., one or more of the vital signs).

(125) In any of these examples, the one or more energies may include a first combination of the plurality of energies **32** (e.g., impact energy **32A** and pressure energy **32D**); and a second combination of the plurality of energies **32** (e.g., heat energy **32B** and pressure energy **32D**). The first combination may be followed by any second combination(s). For example, the one or more energies may include a first energy (e.g., impact energy **32A**) communicable with a first portion of the nerves (e.g., Meissner's corpuscle); and a second energy (e.g., heat energy **32D**) communication with a second portion of the nerves (e.g., the Ruffini corpuscle).

(126) Operating step **1040** may alternatively comprise: operating, with processing unit **60**, the plurality of energy generators **31** to communicate energy signal **90** to nerves associated with the skin **2** by outputting one or more energies of the plurality of energies **32** in signal direction SD based on input data **80**. For example, step **1040** may comprise outputting different combinations of the one or more energies, and each different combination may communicate a different portion of the energy signal **90**. Similar to above, energy signal **90** may include one or more symbols **92**, and operating step **1040** may comprise outputting the one or more energies to communicate the one or more symbols **92**. Step **1040** may comprise scrolling the one or more symbols **92** across skin **2** in a communication direction CD transverse with the signal direction SD; and/or flashing any of symbols **92** on-and-off. The plurality symbols **92** may include any type of signal, including pip patters, alphanumeric symbols, and the like.

(127) Various energy types may be used. For example, operating step **1040** may comprise outputting a first combination of the one or more energies to communicate a first symbol of the one or more symbols (e.g., symbol **92A**), and outputting a second combination of the one or more energies to communicate a second symbol of the one or more symbols (e.g., symbol **92B**). In some

aspects, operating step **1040** may comprise: outputting a first combination of the one or more energies to communicate energy signal **90** and outputting a second combination of the one or more energies to communicate a characteristic of energy signal **90**, so as to highlight energy signal **90** or a portion thereof. Input data **80** may include a measurement, and step **1040** may comprise outputting a first combination of the one or more energies based on the measurement. In this example, step **1040** may comprise modifying the first combination based on a change of the measurement, and/or outputting a second combination of the one or more energies based on the change of the measurement.

(128) The larger size of transceiver **910** relative to transceivers **10** may allow for different methods of operation. As shown in FIG. **10**, for example, an exemplary method **1100** may comprise: receiving, with processing unit **960**, input data **980** for a communication device **910** including a tissue interface **930** maintainable on or adjacent skin **2**, the interface **930** including a plurality of energy generators **931** arranged in bands **924.sub.1** and **924.sub.2**, each generator **931** being operable to output a plurality of energies **32** in a signal direction SD toward the skin **2** (a receiving step **1120**); and operating, with processing unit **60**, the plurality of energy generators **931** in each band **924.sub.1** and **924.sub.2** to communicate with nerves associated with the skin **2** by outputting one or more energies of the plurality of energies **32** in response to the input data **80** (an operating step **1140**).

(129) Receiving step **1020** may comprise receiving input data **980** from one or more data sources **981**. As shown in FIG. **10**, for example, step **1020** may comprise receiving a first input data **980A** from a first data source **981A**, and a second input data **980B** from a second data source **981B**. Input data **980** may include a plurality of measurements. Accordingly, receiving step **1020** may comprise receiving input data include a plurality of measurements; and operating step **1040** may comprise operating the plurality of energy generators **931** in each band **924.sub.1** and/or **924.sub.2** to output the one or more energies based on one measurement of the plurality of measurements.

(130) In the healthcare setting, first data source **981A** may include a patient monitoring device or sensor configured to output measurements associated with a first patient, and second data source **981A B** may include a patient monitoring device or sensor configured to output measurements associated with a second patient. The measurements may include vital signs for the respective first and second patients. In this example, receiving step **1120** may comprise receiving input data **980** including a plurality of vital signs; and operating step **1140** may comprise operating the plurality of energy generators **931** in each band **924.sub.1** and **924.sub.2** to output the one or more energies based on one vital sign of the plurality of vital signs. For example, step **1140** may comprise operating the generators **931** in band **924.sub.1** to output energies **32** based on the vital signs for the first patient, and/or operating the generators **931** in band **924.sub.2** to output energies **32** based on the vital signs for the second patient.

(131) Aspects of energies **32** may be modified based on the measurements. For example, operating step **1040** may comprise: outputting a first combination of energies **32** when the at least one of the measurements is inside of an acceptable range; and outputting a second combination of energies **32** when at least one of the measurements is outside of the acceptable range. In the healthcare setting, one of the vital signs of the patient (e.g., pulse rate) may serve as the baseline measure.

(132) Similar to above, input data **980** may include a control signal for each band **924.sub.1** and **924.sub.2**, and operating step **1140** may comprise outputting the energies **32** based on the control signal for each band **924.sub.1** or **924.sub.2**. Alternatively, method **1100** may further comprise: generating, with the processing unit **960**, a control signal for each band **924.sub.1** and **924.sub.2** based on input data **980**, wherein the operating step **1140** comprises outputting the energies **32** based on the control signal for each band.

(133) Also similar to above, operating step **1140** also may comprise operating the plurality of energy generators **931** to simultaneously communicate a plurality of energy signals to nerves associated with the skin **2** by outputting an energy signal in each band with energies **32**, and/or

scrolling the energy signal in its respective band. As shown in FIG. 10, step 1140 may comprise outputting first energy signal 990.sub.1 in first band 924.sub.1 with a first combination of energies 32, and outputting second energy signal 990.sub.2 in second band 924.sub.2 with a second combination of energies 32. Each signal 990.sub.1 and 990.sub.2 may include a plurality of symbols (e.g., symbols 92), and operating step 1140 may comprise scrolling the symbols across one of bands 924.sub.1 and 924.sub.2. In keeping with above, first signal 990.sub.1 (and any symbols contained therein) may be scrolled along a first communication direction CD.sub.1 transverse with signal direction SD, and second signal 990.sub.2 (and any symbols contained therein) may be scrolled along a second communication direction CD.sub.2 transverse with signal direction SD.

(134) Although described with reference to two divided areas (e.g., first band 924.sub.1 and second band 924.sub.2) configured to output two energy signals (e.g., first signal 990.sub.1 and second signal 990.sub.2), it is contemplated that method 1100 may be configured for any number of divided areas. Accordingly, variations of method 1100 may allow the user to simultaneously monitor a plurality of sources of input data, from one or more data sources, with aspects of transceiver 910 described herein.

(135) Additional aspects described above with reference to transceivers 10, 110, 210, 310, 410, 510, 610, 710, 810 and 910, and methods 1000 and 1100, are now described with reference to a communication system 2000. Aspects of an exemplary system 2000 are depicted in FIGS. 11 and 12. As shown in FIG. 11, communication system 2000 may comprise a plurality of energy transceivers configured to receive input data and output one or more of a plurality of energies 32 to different locations of skin 2 according to the input data. Each transceiver may include any element and perform any function described above with reference to transceivers 10, 110, 210, 310, 410, 510, 610, 710, 810 and 910, and methods 1000 and 1100. Different aspects may be combined in system 2000. For example, as shown in FIG. 11, system 2000 may comprise: a first energy transceiver 2012 on a head of a user 1, similar to transceiver 110 of FIG. 6A; a second energy transceiver 2014 on one arm of user 1, similar to transceiver 900 of FIGS. 9A-B; a plurality of energy transceivers 2016 attached to a torso of user 1, similar to transceiver 10 of FIGS. 2A-C; an energy transceiver 2018 on each leg of user 1, similar to transceiver 410 of FIG. 6D; and an energy transceiver 2020 in each shoe of user 1, similar to transceiver 510 of FIG. 7A.

(136) Each transceiver 2012, 2014, 2016, 2018, and 2020 may be interconnected in system 2000 so that coordinated control signals may be output to each transceiver for output of a corresponding signal 90 with a corresponding one or more of energies 32. The coordinated control signals may be used to coordinate activities or movements of user 1 in response to the input data. As shown in FIG. 11, each of said transceivers may output data associated with a first form or position of user 1, receive input data regarding a second form or position, and output signals 90 in the same or communication directions CD to direct the user 1 to move according to the desired form or position. In one aspect, the first form or position may be a first pose or stance, and the second form or position may be a second pose; in other aspects, the first form or position may be a first (e.g., GPS) position on a field, and the second form or position may be a second (e.g., GPS) position on the field.

(137) As shown in FIG. 11, for example, transceiver 2012 may scroll a first signal 90 in a first communication direction CD.sub.1 around the head; transceiver 2014 may scroll a second signal 90 around the arm in a second communication direction CD.sub.2; transceivers 2016 may output third signals 90 without scrolling; transceivers 2018 may scroll fourth signals 90 in communication directions CD.sub.3R and CD.sub.3L around the legs; transceivers 2020 may scroll fourth signals 90 in communication directions CD.sub.4R and CD.sub.4L across the feet. Accordingly, each of the respective signals and communication directions may be coordinated in system 2000 to direct the user to move in a particular direction and/or move one or more of their limbs into a particular form or position.

(138) Aspects of methods **1000** and **1100** may be modified for use within system **2000**. As shown in FIG. **12**, for example, an exemplary method **2100** may comprise: receiving, with one or more processors, position data for a plurality of communication devices (e.g., transceivers **2012**, **2014**, **2016**, **2018**, **2020**) mountable on or adjacent skin, each device including a tissue interface with a plurality of energy generators **31**, each generator **31** being operable to output a plurality of energies **32** in a signal direction SD toward skin **2** (a “receiving step **2120**”); receiving or generating, with the one or more processors, a corrective motion signal for the plurality of communication devices based on position data for each communication device (a “receiving or generating step **2140**”); and operating, with the one or more processors, the plurality of energy generators **31** of each communication device to output one or more energies of the plurality of energies **32** in signal direction SD based on the corrective motion signal. Although described with reference to elements of transceiver **10**, method **2100** may be performed with any transceiver described herein.

(139) Additional aspects described above with reference to transceivers **10**, **110**, **210**, **310**, **410**, **510**, **610**, **710**, **810** and **910**, methods **1000** and **1100**, system **2000**, and method **2100** are now described with reference to aspects of an energy transceiver **3010** shown in FIGS. **13A**, **13B**, **14A**, and **14B**. As before, any aspect of energy transceiver **3010** may be combined with any aspect described above.

(140) As shown in FIG. **13A**, energy transceiver **3010** may include: a body **3020** and a tissue interface **3030**; and an attachment element **3070**, shown conceptually as a band in this example. As above, body **3020** may wrap around a circular portion of skin **2**, such as around the human forearm shown in FIG. **13B**. For example, as before, body **3020** may be mounted on attachment element **3070**; and tissue interface **3030** may be mounted on a distal surface of body **3020**, providing a curved rectangular communication area **4** and a semi-circular (e.g., less than 360°) or circular (e.g., 360°) communication direction CD for energy signal **90**. In keeping with above, attachment element **3070** (e.g., the band) may be configured to maintain tissue interface **3030** against or the forearm when element **3070** is worn, allowing energy signal **90** to be output communication area **4** in signal direction SD and/or scrolled across area **4** in communication direction CD.

(141) As described above, aspects of each energy **32** may be modified to increase the complexity of energy signal **90**, and thus the amount of data transmitted therewith; and the modifiable aspects may include energy type, energy intensity, output duration, scroll rate, symbol shape, and the like, providing an incredibly broad range of obtainable complexity. Training may be required to leverage the full communicative capabilities of tissue interface **3030** and signal **90**. For example, within a repetition program, the user may be trained to more easily and/or quickly to distinguish between: any number of known shapes output by one of energies **32**, such as between a pip two dot pattern output with impact energy **32A** and a pip four dot pattern output with energy **32A**; or the same shape output with different energies **32**, such as a pip five dot pattern with impact energy **32A** or heat energy **32B**.

(142) Communicating more complex variations, unknown signals, and/or unknown shapes may require additional training. For example, interface **3030** may output energy signal **90** to include pip patterns in which each dot is output with a different combination of energies **32**, allowing the pattern to be associated with a target, and each dot to be associated with a characteristic thereof. In the healthcare setting, for example, the pattern may be associated with a patient, and each dot may be associated with a different vital sign of the patient, providing immediate insight into patient health that may be updated continuously. Further training may be required to quickly distinguish between the characteristics communicated by each dot in these examples, particularly if energy signal **90** includes a plurality of pip patterns, as shown in FIG. **2C**; or a dynamic shape, such as the echocardiogram depicted in FIGS. **13A** and **13B**; the plurality of echocardiograms depicted in FIG. **14A**; or the alphanumeric symbol stream depicted in FIG. **14B**.

(143) Aspects of energy transceiver **3010** may be configured to provide additional communicative capabilities to, for example, assist with training. As shown in FIG. **13A**, transceiver **3010** may

further comprise an optical interface **3030'** compatible with eyes of the user. For example, optical interface **3030'** may comprise at least one display element operable to output an optical energy signal **90'** to the eyes, such as a flexible LED configured to output a plurality of colors. Any display technology may be used. As shown in FIG. **13A**, interface **3030'** may provide a curved optical communication area that wraps around apparatus **3010** along an axis X-X and/or substantially corresponds with the communication area **4**. For example, tissue interface **330** may be configured to output non-optical energy signal **90** toward skin **2** with one or more energies **32** in a first or distal direction toward skin **2**; and optical interface **330'** may be configured to output optical signal **90'** with one or more colors in a second or proximal direction toward the eyes.

(144) Energy transceiver **3010** may comprise a processing unit similar to any variation of processing unit **60** described herein. For example, the processing unit may be operable with tissue interface **3030** and optical interface **3030'** to simultaneously communicate with nerves associated with skin **2** and the eyes by outputting signal **90** distally and signal **90'** proximally at the same time. Additional training capabilities may be realized by the simultaneous outputs. For example, the user may already be trained to react to optical signal **90'**, whether or not signal **90** is communicated, such as when transceiver **3010** excludes interface **3030**. Accordingly, by consistently outputting energy signal **90** with optical signal **90'**, the user may be trained to react to recognize and react to energy signal **90** with or without optical signal **90'**.

(145) In a healthcare setting, for example, optical signal **90'** may communicate a vital sign of a patient to the eyes of a provider, such as the echocardiogram of FIG. **13A**; and energy signal **90'** may communicate the same vital sign to skin **2** of the provider at the same time. For example, signals **90'** and **90** may be scrolled together in communication direction CD along axis X-X to simultaneously communicate aspects of the vital sign over time. As a further example, signal **90'** may comprise a plurality of colors, and the output of energies **32** in signal **90** may be modified according to a color matching algorithm to communicate similar aspects to skin **2** at the same time. Reactions to different vital signs may be trained in this manner. As shown in FIG. **13B**, for example, a first portion of optical interface **3030'** may output a first optical signal **90A'**, a second portion of interface **3030'** may output a second optical signal **90B'**, corresponding portions of tissue interface **3030** may output corresponding energy signals **90**, much like interface **930** described above. As also shown in FIG. **13B**, the signals **90A'** and **90B'** may be different, in which one may be a vital sign and other may include symbols communicating related patient data as above.

(146) Accordingly, by simultaneously outputting optical signal **90'** together with energy signal **90**, transceiver **3010** may train reactions to any stimulus, such as the exemplary vital signs and signals depicted in FIGS. **13** and **13B**. As shown in FIGS. **14A** and **14B**, the complexity of the stimulus may be increased. For example, as shown in FIG. **14A**, optical interface **3030'** and tissue interface **3030** may output their respective signals in a plurality of rows arranged around axis X-X, wherein each row includes a different set of corresponding signals movable along a communication direction CD that is transverse with axis X-X. In this example, four rows are shown as outputting four different optical signals, including a first optical signal **90A'**, a second optical signal **90B'**, a third optical signal **90C'**, and a fourth optical signal **90D'**. A corresponding set of rows and outputs may be realized by tissue interface **90**.

(147) In the healthcare setting, for example, each output of optical signals **90A'**, **90B'**, **90C'** and **90D'** together with its corresponding energy signal **90** may communicate a different vital sign of a different patient to a provider, training them to simultaneously monitor all of the different patients at once. As described above, aspects of each energy signal **90**, such as energies **32**, may be modified to communicate changes in the associated vital sign. For training purposes, the color of optical signals **90A**, **90B**, **90C**, and **90D** may be varied based on these changes so that the provider may be trained to first recognize the changes based one of the optical signals; and second recognize the same changes based on one of the energy signals based on the color matching algorithm. For example, the color matching algorithm may comprise a correspondence between visual colors and

energy intensity, in which warmer colors (e.g., red) are associated with higher intensities and cooler colors (e.g., blue) are associated with lower intensities.

(148) Another example is provided in FIG. 14B, in which each output of signals **90A'**, **90B'**, **90C'** and **90D'** together with its corresponding signal **90** may communicate aspects of an alphanumeric stream. As shown in FIG. 14B, for example, each alphanumeric stream may comprise a stock ticker so that the user may be trained to simultaneously monitor a plurality of tickers. As before, aspects of the different optical signals **90A'**, **90B'**, **90C'**, and **90D'** may be modified simultaneously with aspects of their corresponding energy signals **90** to communicate changes over time.

(149) In keeping with above, optical interface **3030'** and tissue interface **3030** may be configured to individually and/or simultaneously output signals **90'** and **90** to include any symbols and shapes, as well as more complex depictions, such as graphics. For example, for more complex depictions, the color matching algorithm may be used to output different combinations of energies **32** based on color.

(150) Optical interface **3030'** may comprise touchscreen capabilities allowing manipulation of signals **90** and/or **90'** by interaction therewith. For example, the position of each row depicted in FIGS. 14A and 14B may be movable via a tactile interaction with interface **3030'**. As shown in FIG. 13B, for example, attachment element **3070** may maintain the position of tissue interface **3030** on or adjacent skin **2** of a forearm, meaning that at least some portion of optical interface **3030'** may not be aligned with the eyes of the user at all times. Accordingly, because of the dynamic capabilities of interfaces **3030** and **3030'**, the touchscreen capabilities of apparatus **3010** may allow the user to move a particular row into alignment with the eyes by scrolling the rows together around axis X-X, in which the outputs of signals **90A'**, **90B'**, **90C'**, and **90D'** and corresponding energy signal **90** move with each row. Any type of touchscreen-enabled two-way communication means may be used, including buttons, sliders, textual inputs, graphic inputs, and the like.

(151) Aspects of methods **1000**, **1100**, and **2100** and/or system **200** may be modified according to aspects of energy transceiver **3010**. For example, any method steps described herein may be modified to comprise training and/or communication steps according to the above-described aspects of transceiver **3010**. As a further example, the second energy transceiver **2014** shown in FIG. 11 may comprise transceiver **3010**, which may be further operable with each of transceivers **2012**, **2016**, **2018**, and **2020** to train the user. To provide another example, aspects of each transceiver within system **2000** also may be configured to placement at a particular sensory zone of skin **2**, and transceiver **3010** may be used to both tune the respective energy signals **90** for output to each zone and train the user to react accordingly based on one or more of the signals **90**. In this example, the receptive capabilities of the nerves associated with skin **2** in each zone may vary, and transceiver **3010** may be configured to operate the transceivers in system **2000** so that the most complex signals are communicated to the most receptive zones.

(152) While principles of the present disclosure are disclosed herein with reference to illustrative aspects for particular applications, the disclosure is not limited thereto. Those having ordinary skill in the art and access to the teachings provided herein will recognize additional modifications, applications, aspects, and substitution of equivalents all fall in the scope of the aspects disclosed herein. Accordingly, the present disclosure is not to be considered as limited by the foregoing description.

Claims

1. A device, comprising: a body housing a power source, a processing unit, and a bus including an electrical connector; and a generator configured to be releasably secured to the body, the generator comprising at least part of a first energy generator element and a second energy generator element, the first energy generator element and the second energy generator element being independently operable to convert electricity into a first energy type and a second energy type, respectively, and to

communicate the first energy type and the second energy type toward an area of skin, wherein the generator further includes one or more electrical conductors engageable with the electrical connector when the generator is secured to the body to provide at least one electrical pathway from the power source to the generator, wherein the body includes a grip arranged to be grasped by a hand of a user applying a gripping force to maintain the first energy generator element and the second energy generator element on or adjacent the area of skin, wherein the first energy generator element is an impact generator element and includes a drive mechanism, a piston, and a tissue contact surface that is linearly actuatable along an axis to contact and cause corresponding physical movement of the area of skin, and wherein the second energy generator element is arranged coaxially about the axis.

2. The device of claim 1, further comprising a resilient element provided between the drive mechanism and the tissue contact surface.

3. The device of claim 1, wherein the drive mechanism is a motor, and wherein the first energy generator element is operably coupled to a controller that directs the electricity to the motor to move the piston and the tissue contact surface along the axis.

4. The device of claim 1, wherein the one or more electrical conductors comprises at least one electrical wire.

5. The device of claim 4, wherein at least a portion of the one or more electrical conductors extends substantially parallel to the axis.

6. The device of claim 4, wherein the at least one electrical wire is twisted.

7. The device of claim 4, wherein the at least one electrical wire establishes an electrical pathway to the second energy generator element.

8. The device of claim 7, wherein the second energy generator element includes a heat generator element.

9. The device of claim 8, wherein the heat generator element includes a conductor and a resistor.

10. The device of claim 1, further comprising a coil, and wherein a portion of the impact generator element is located within an interior of the coil.

11. A device, comprising: a body housing a power source, a processing unit, and a bus including an electrical connector; a generator configured to be releasably secured to the body, the generator comprising at least part of an impact generator element and a heat generator element, the impact generator element and the heat generator element being independently operable to convert electricity into an impact energy and a heat flux, respectively, and to communicate the impact energy and the heat flux toward an area of skin, wherein the impact generator element includes a drive mechanism, a piston, and a tissue contact surface that is linearly actuatable along an axis to contact and cause corresponding physical movement of the area of skin; and a coil, wherein a portion the impact generator element is located within an interior of the coil, wherein the generator further includes one or more electrical conductors engageable with the electrical connector when the generator is secured to the body to provide at least one electrical pathway from the power source to the generator, and wherein the heat energy generator element is arranged coaxially about the axis.

12. The device of claim 11, wherein the drive mechanism is a motor, and wherein the impact generator element is operably coupled to a controller that directs the electricity to the motor to move the piston and the tissue contact surface along the axis.

13. The device of claim 11, wherein the one or more electrical conductors comprises at least one electrical wire.

14. The device of claim 13, wherein at least a portion of the one or more electrical conductors extends substantially parallel to the axis.

15. The device of claim 13, wherein the at least one electrical wire is twisted.

16. The device of claim 15, wherein the at least one electrical wire establishes an electrical pathway to the heat generator element.

17. The device of claim 11, wherein the heat generator element includes a conductor and a resistor.
18. The device of claim 11, wherein the generator includes a housing having an attachment feature comprising a mechanical connector for releasably securing the generator to the body.
19. A device, comprising: a body housing a power source, a processing unit, and a bus including an electrical connector; a generator configured to be releasably secured to the body, the generator comprising at least part of an impact generator element and a heat generator element, the impact generator element and the heat generator element being independently operable to convert electricity into an impact energy and a heat flux, respectively, and to communicate the impact energy and the heat flux toward an area of skin, wherein the impact generator element includes a drive mechanism, a piston, and a tissue contact surface that is linearly actuatable along an axis to contact and cause corresponding physical movement of the area of skin; and a resilient element provided between the drive mechanism and the tissue contact surface, wherein the generator further includes one or more electrical conductors engageable with the electrical connector when the generator is secured to the body to provide at least one electrical pathway from the power source to the generator, and wherein the heat energy generator element is arranged coaxially about the axis.
20. The device of claim 19, wherein at least a portion of the one or more electrical conductors comprises at least one electrical wire, and wherein the at least one electrical wire is twisted.
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